

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

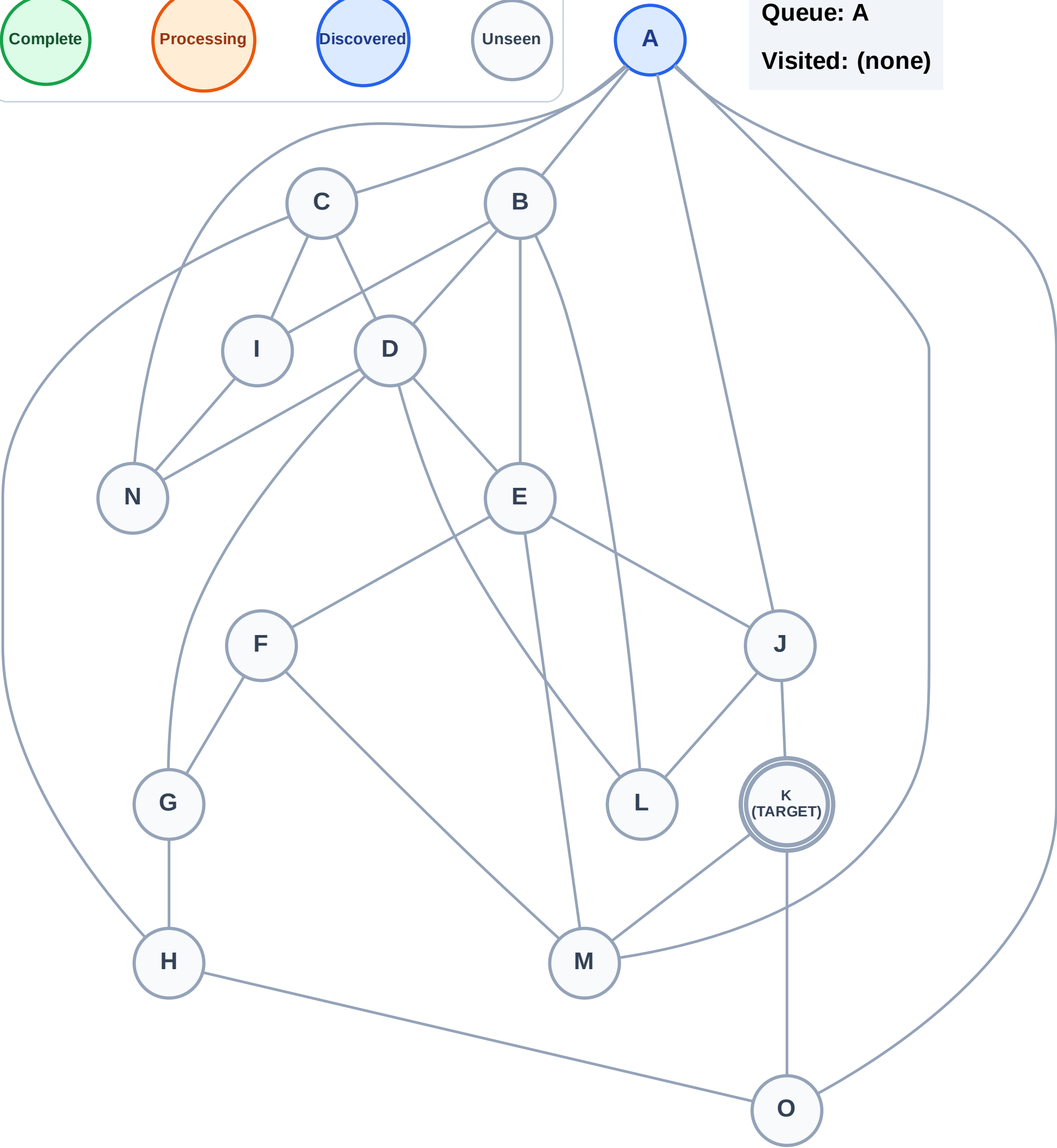
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

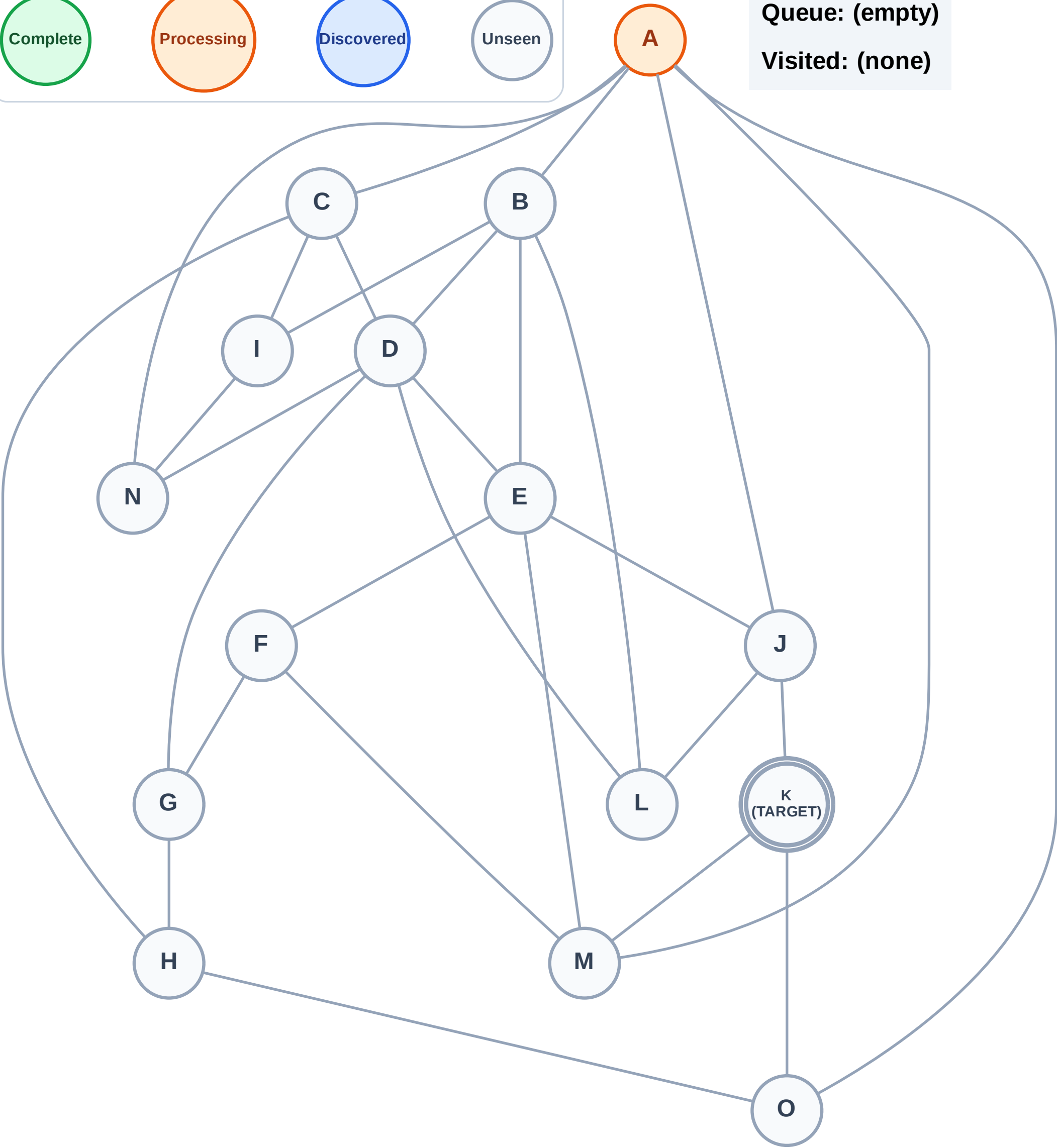
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



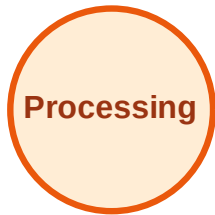
BFS Traversal

Step 3: Discover B, C, J, M, N, O from A

Legend



Complete



Processing



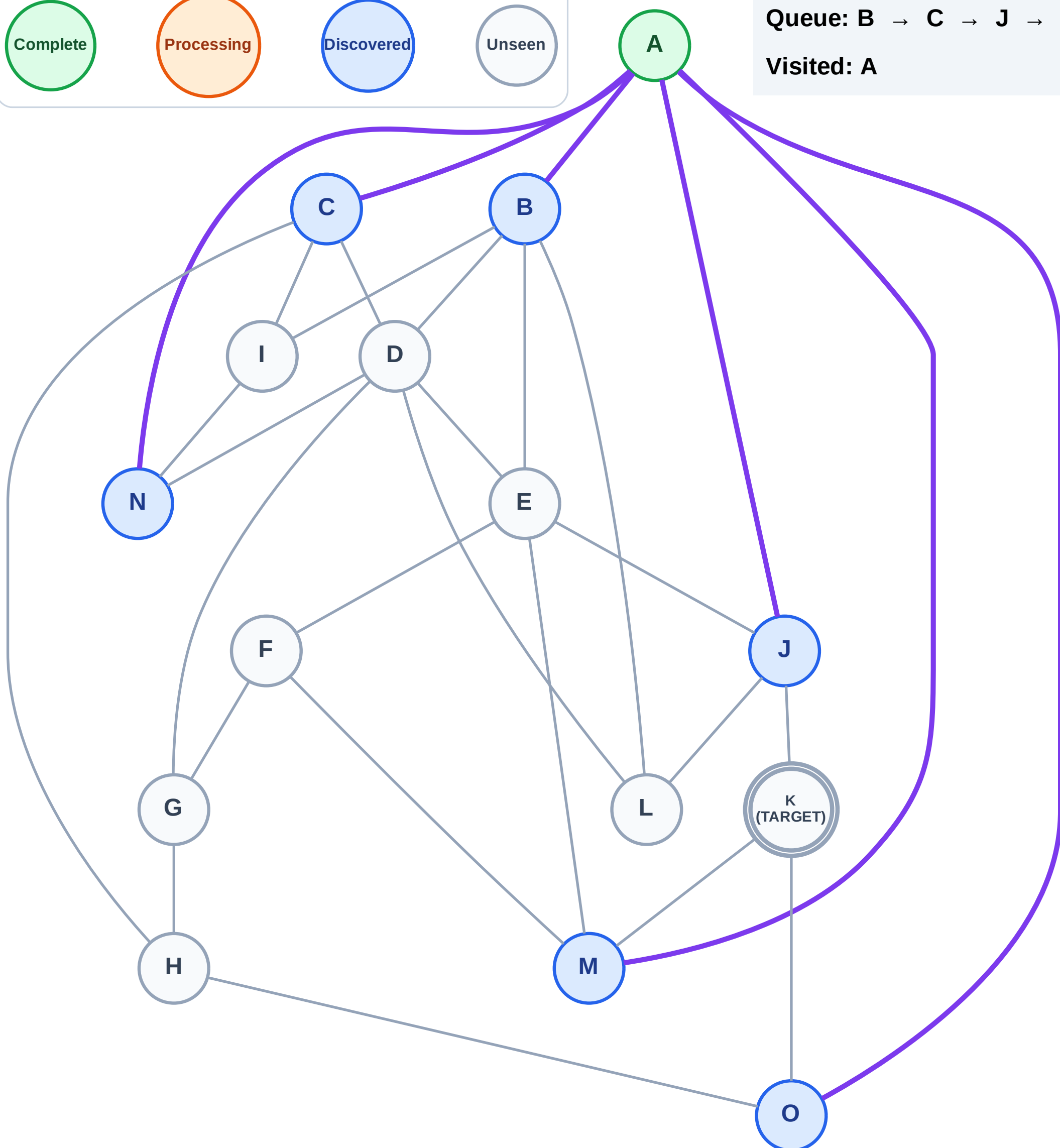
Discovered



Unseen

Queue: B → C → J → M → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



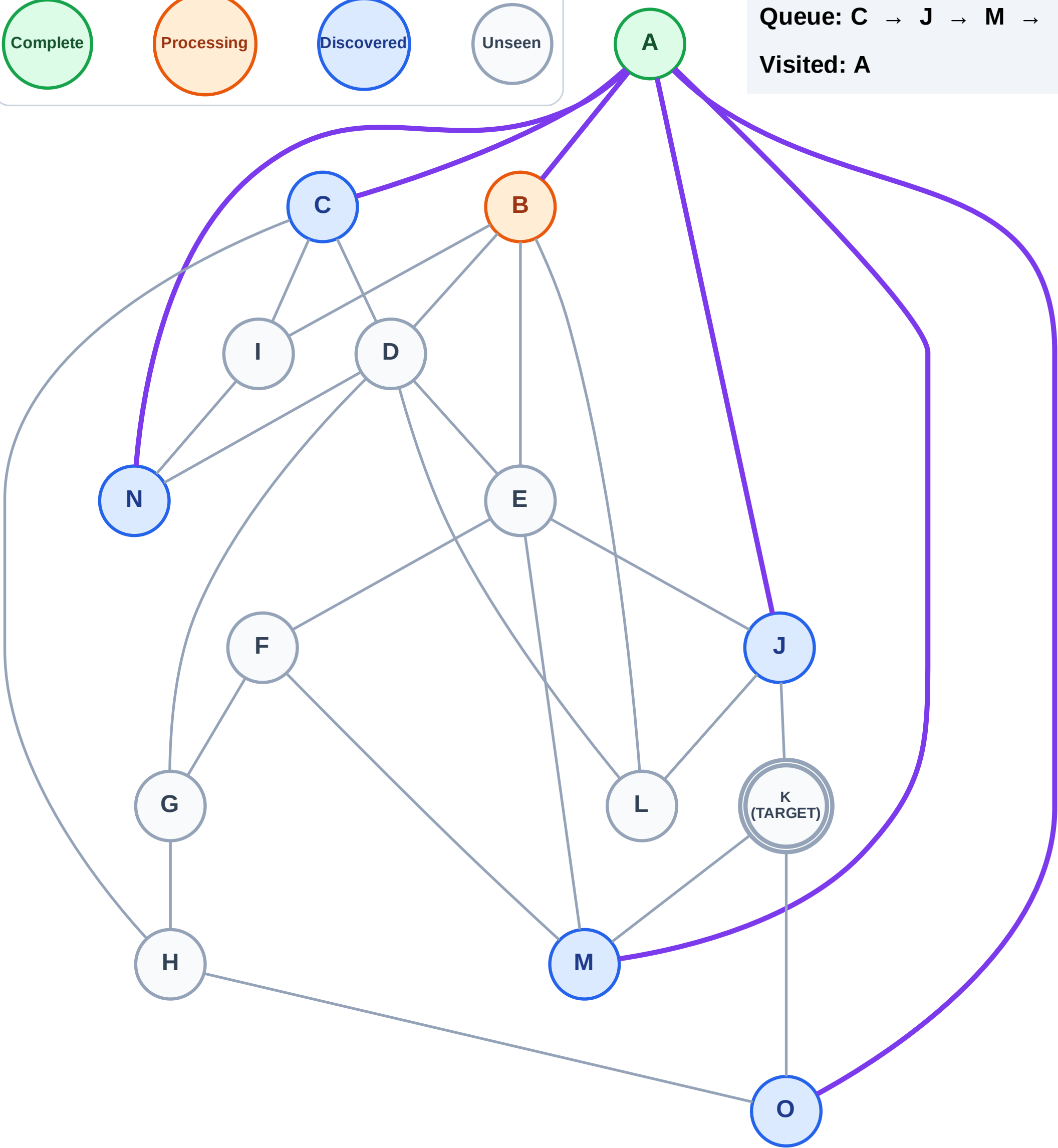
Discovered



Unseen

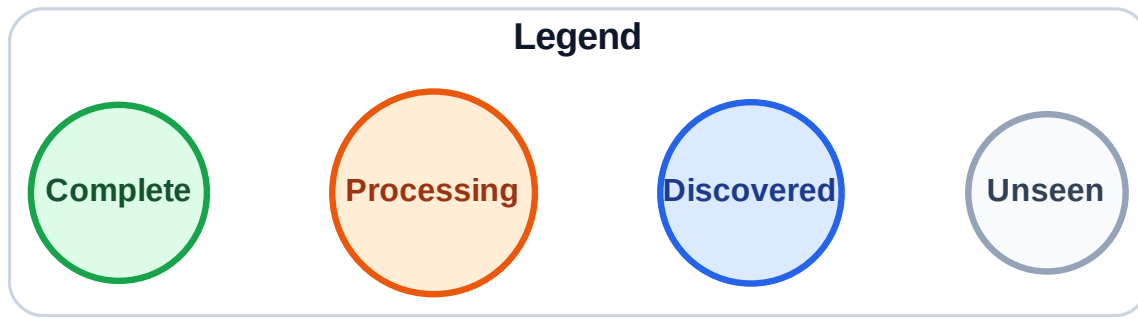
Queue: C → J → M → N → O

Visited: A



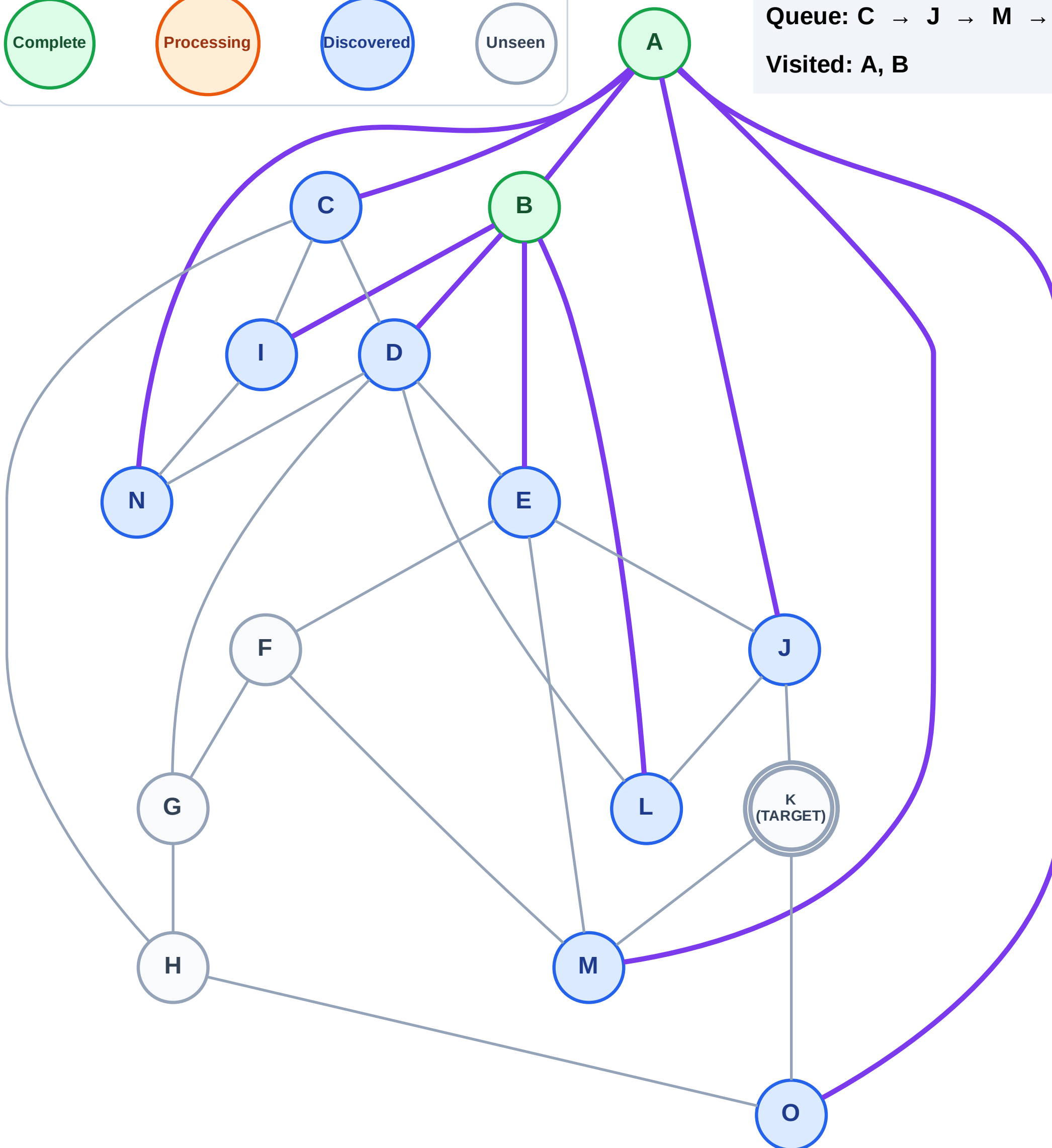
BFS Traversal

Step 5: Discover D, E, I, L from B



Queue: C → J → M → N → O → D → E → I → L

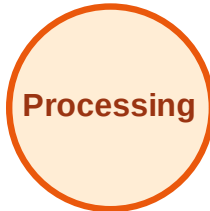
Visited: A, B



BFS Traversal

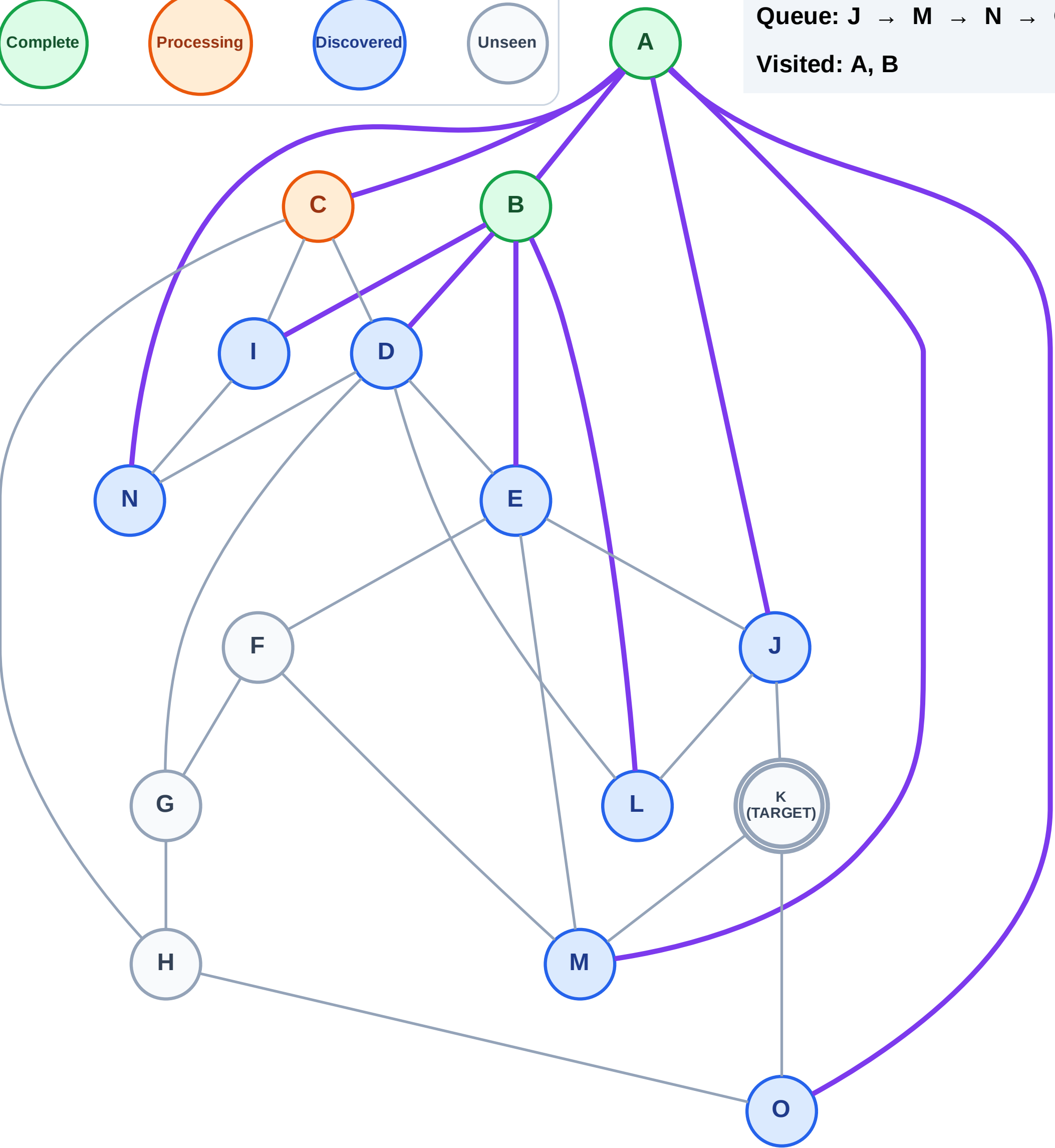
Step 6: Process node C

Legend



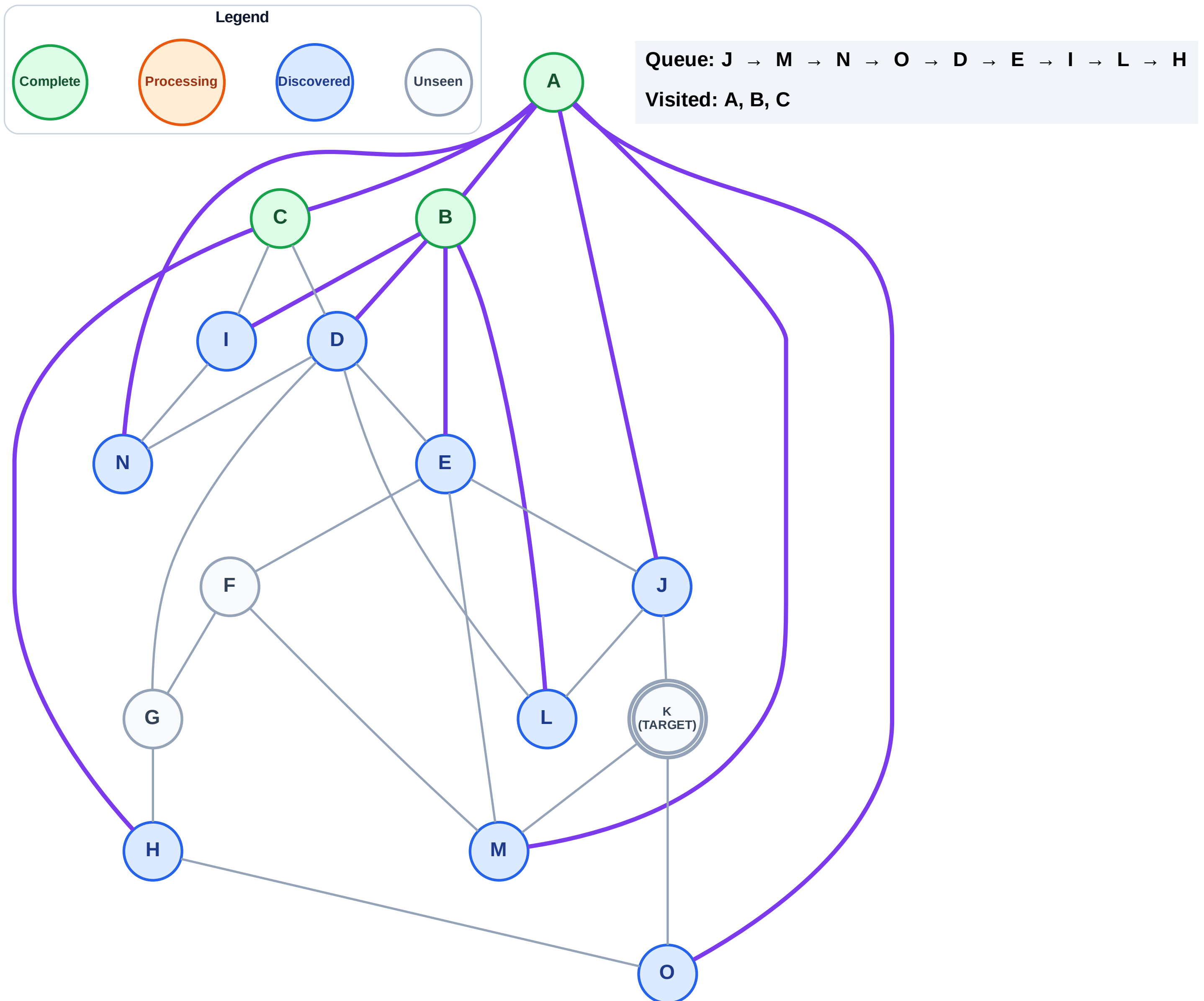
Queue: J → M → N → O → D → E → I → L

Visited: A, B



Step 7: Discover H from C

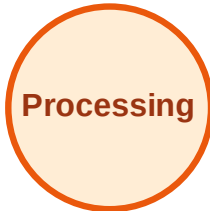
Step 7: Discover H from C



BFS Traversal

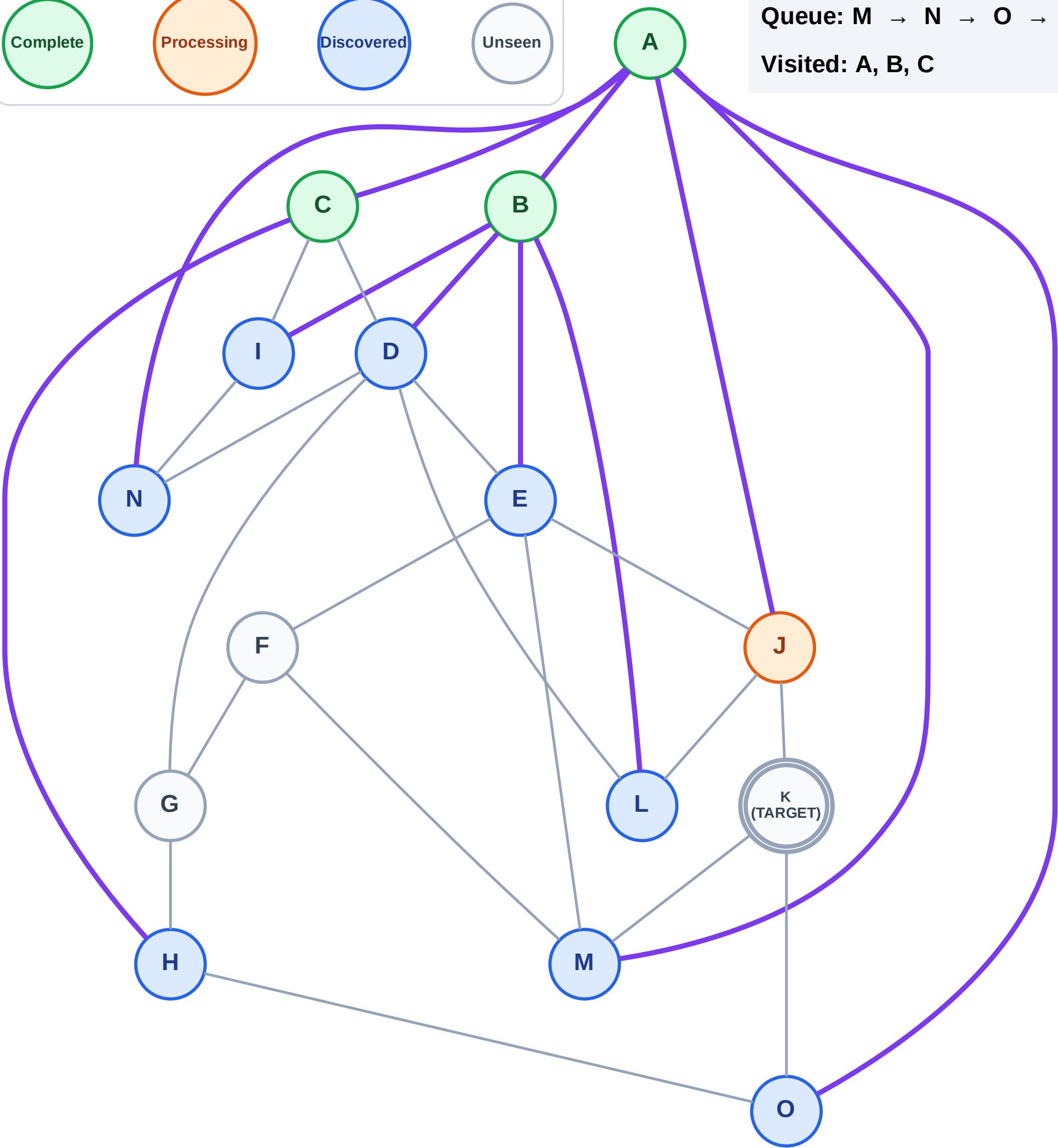
Step 8: Process node J

Legend



Queue: M → N → O → D → E → I → L → H

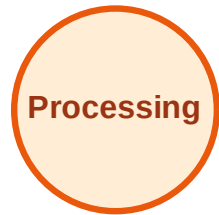
Visited: A, B, C



BFS Traversal

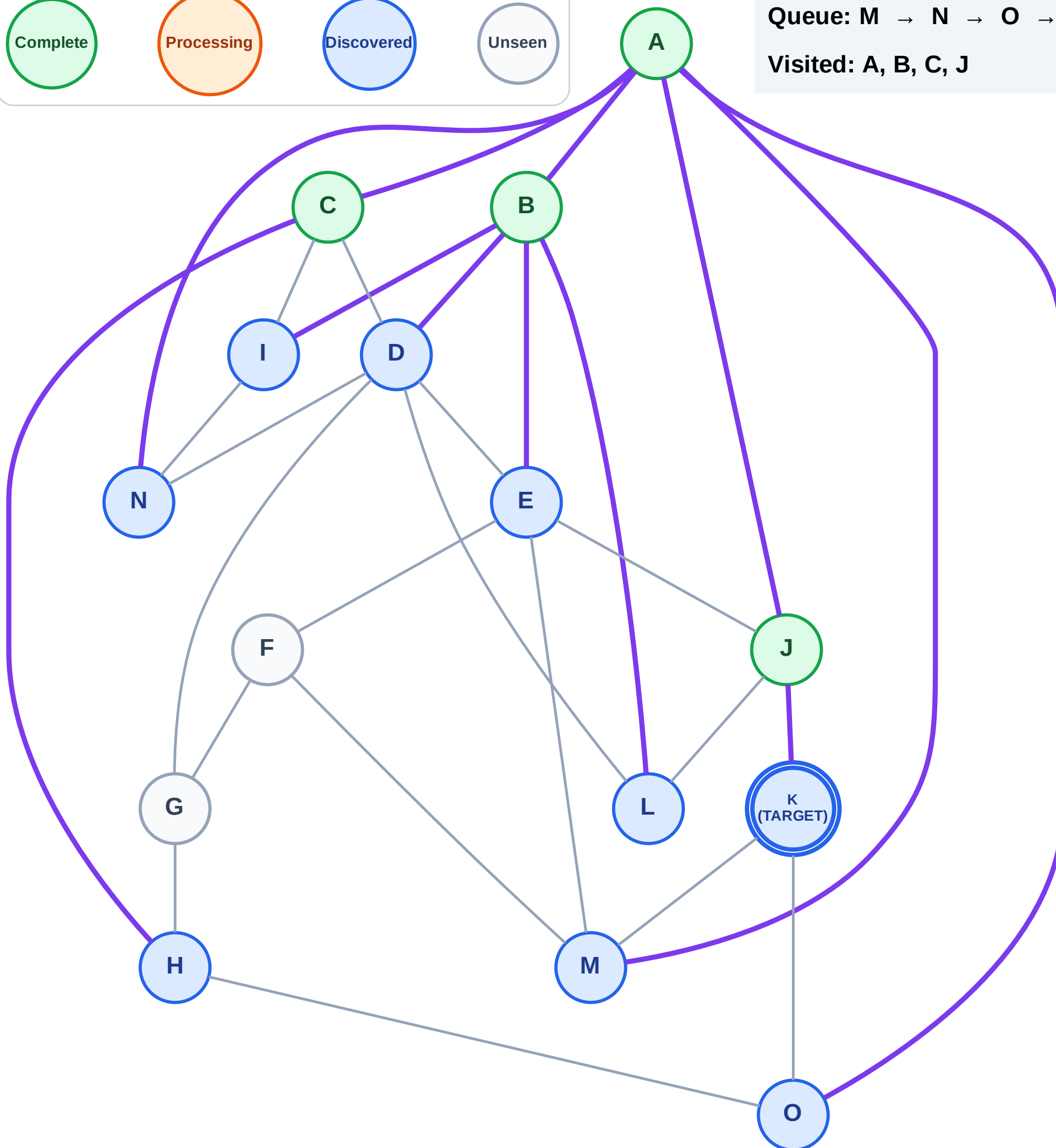
Step 9: Discover K from J

Legend



Queue: M → N → O → D → E → I → L → H → K

Visited: A, B, C, J



BFS Traversal

Step 10: Process node M

Legend

Complete

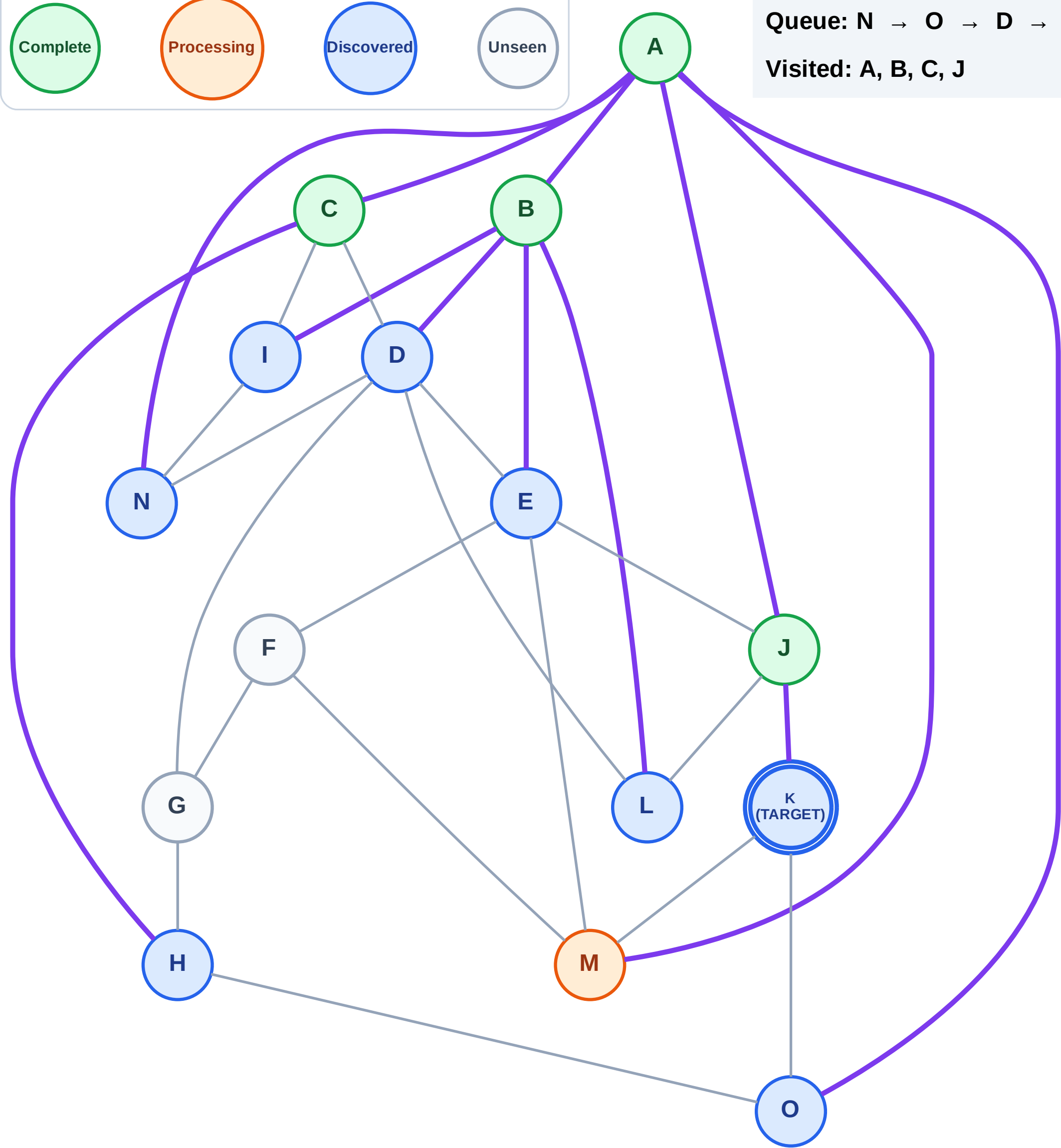
Processing

Discovered

Unseen

Queue: N → O → D → E → I → L → H → K

Visited: A, B, C, J



BFS Traversal

Step 11: Discover F from M

Legend

Complete

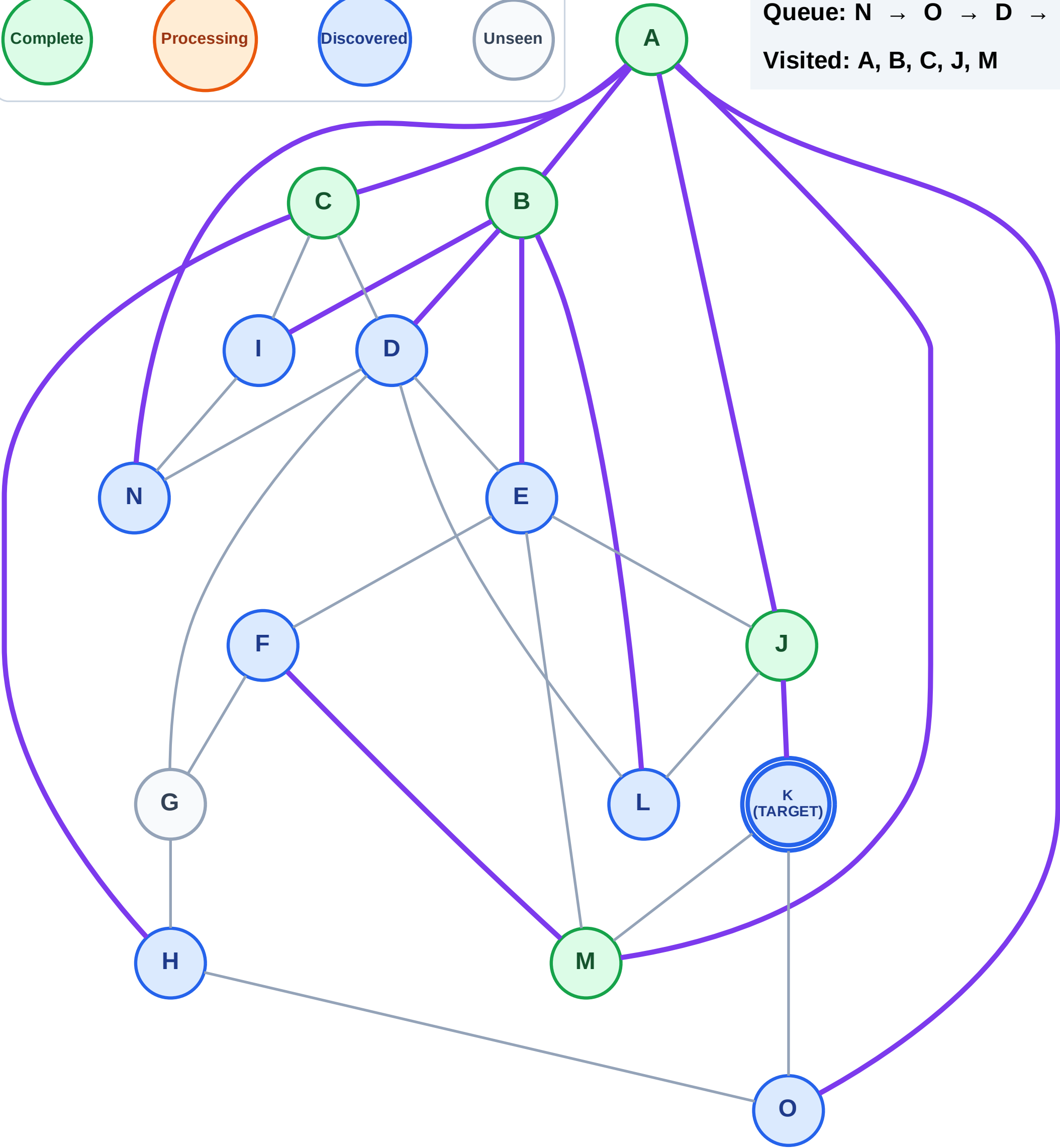
Processing

Discovered

Unseen

Queue: N → O → D → E → I → L → H → K → F

Visited: A, B, C, J, M



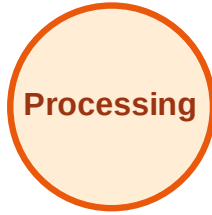
BFS Traversal

Step 12: Process node N

Legend



Complete



Processing



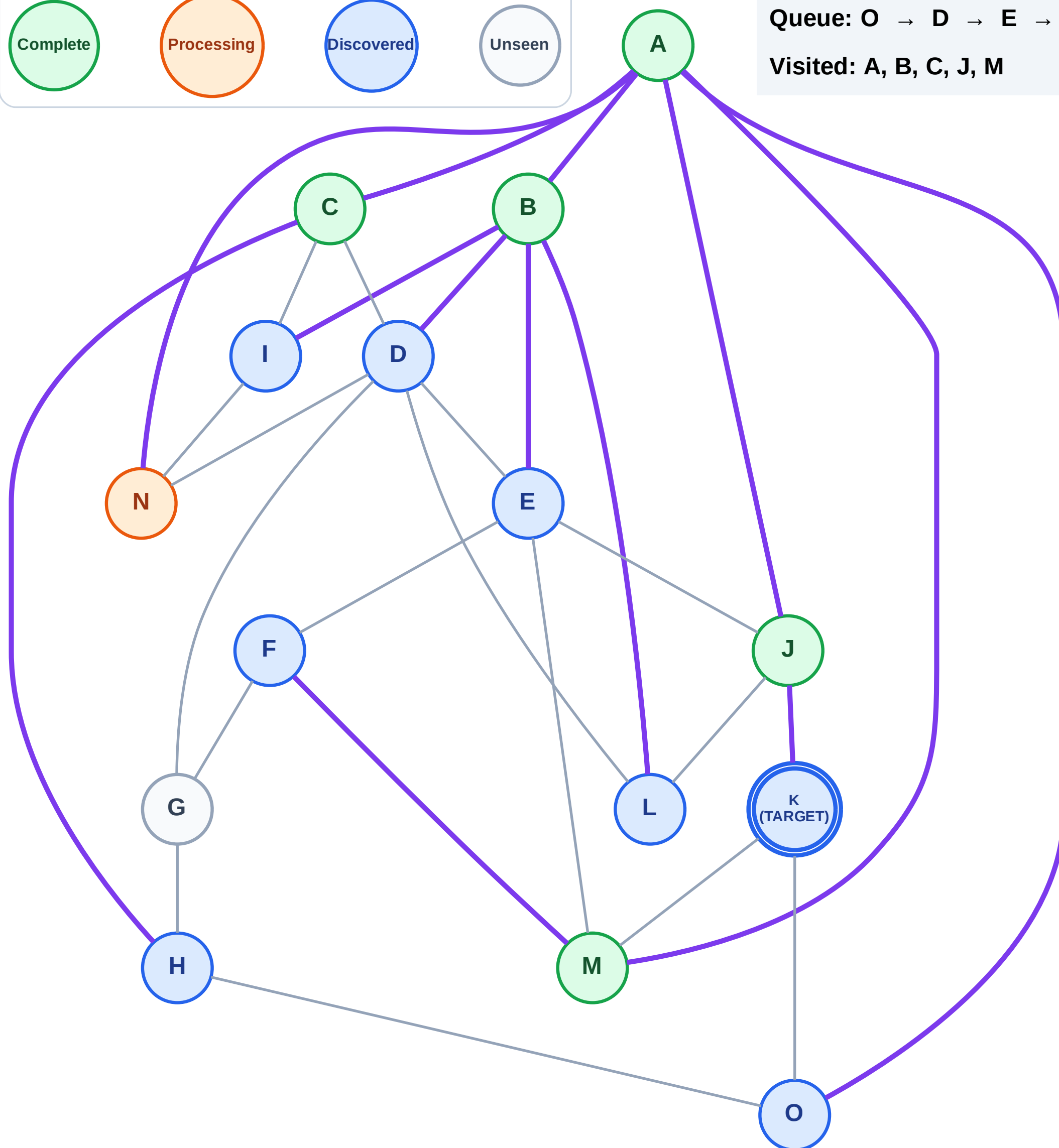
Discovered



Unseen

Queue: O → D → E → I → L → H → K → F

Visited: A, B, C, J, M



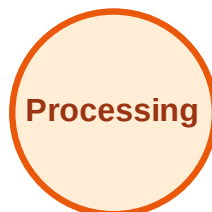
BFS Traversal

Step 13: No new neighbors from N

Legend



Complete



Processing



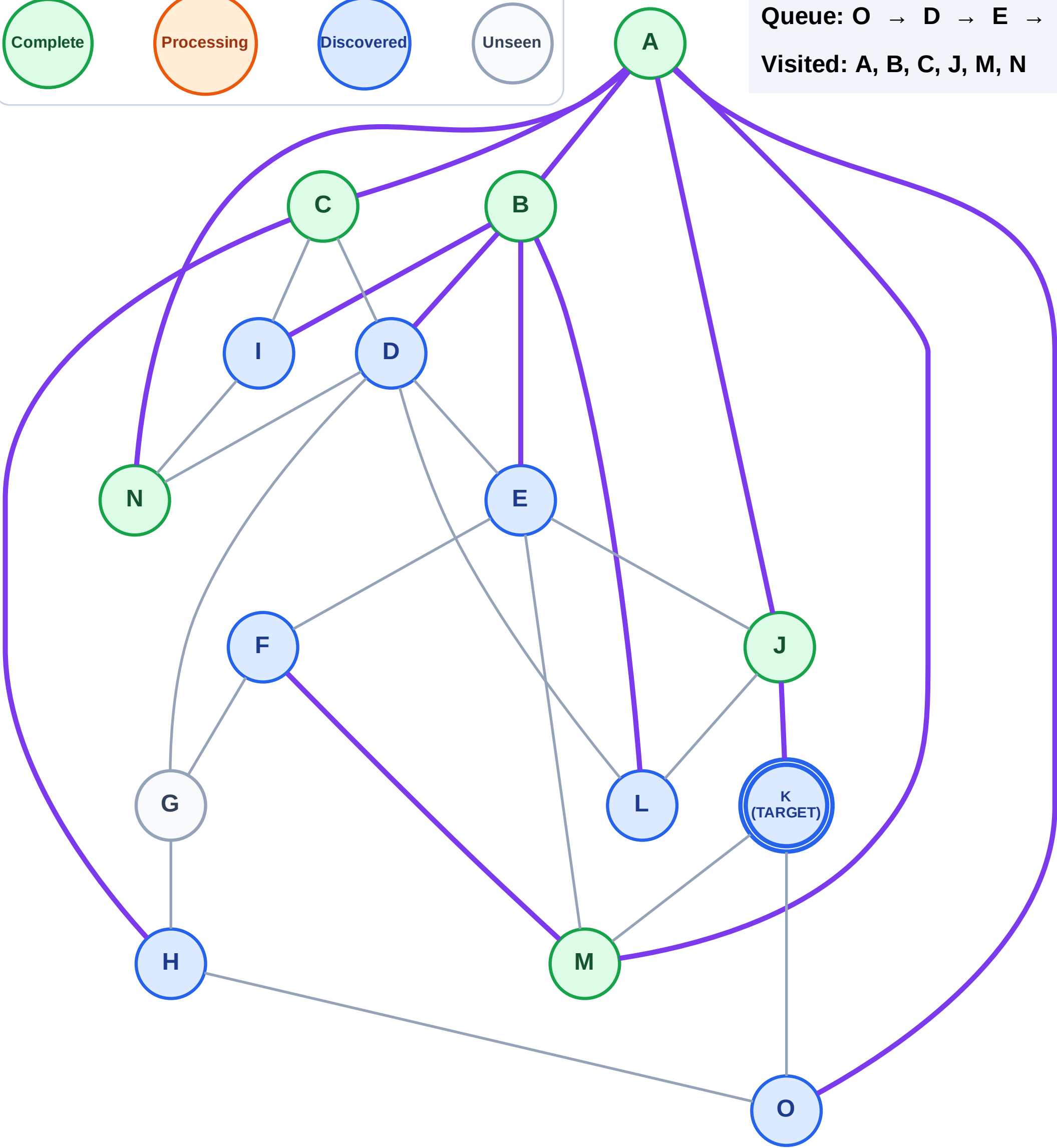
Discovered



Unseen

Queue: O → D → E → I → L → H → K → F

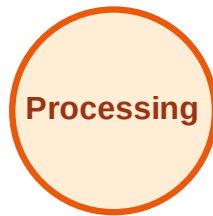
Visited: A, B, C, J, M, N



BFS Traversal

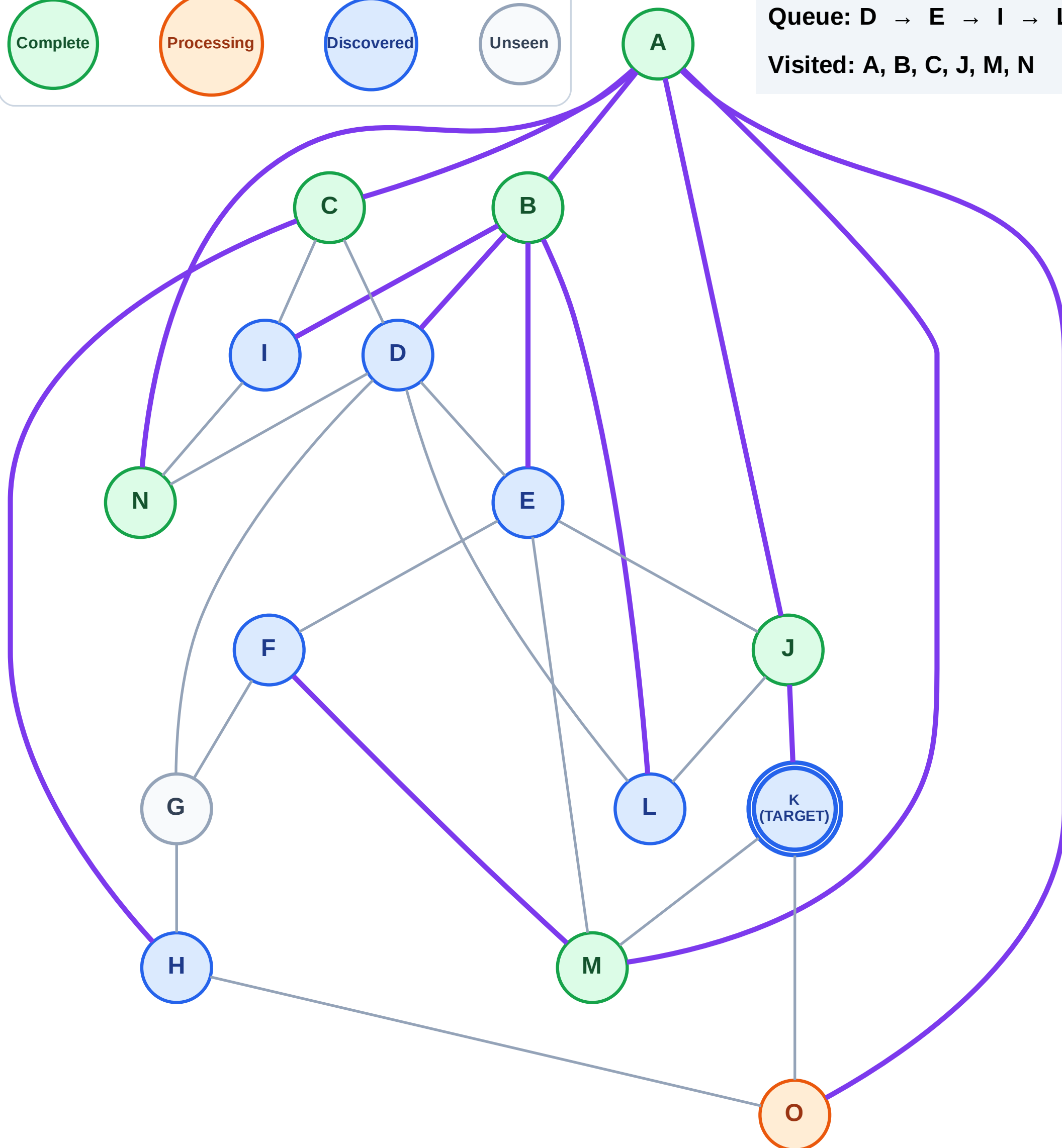
Step 14: Process node O

Legend



Queue: D → E → I → L → H → K → F

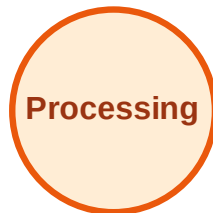
Visited: A, B, C, J, M, N



BFS Traversal

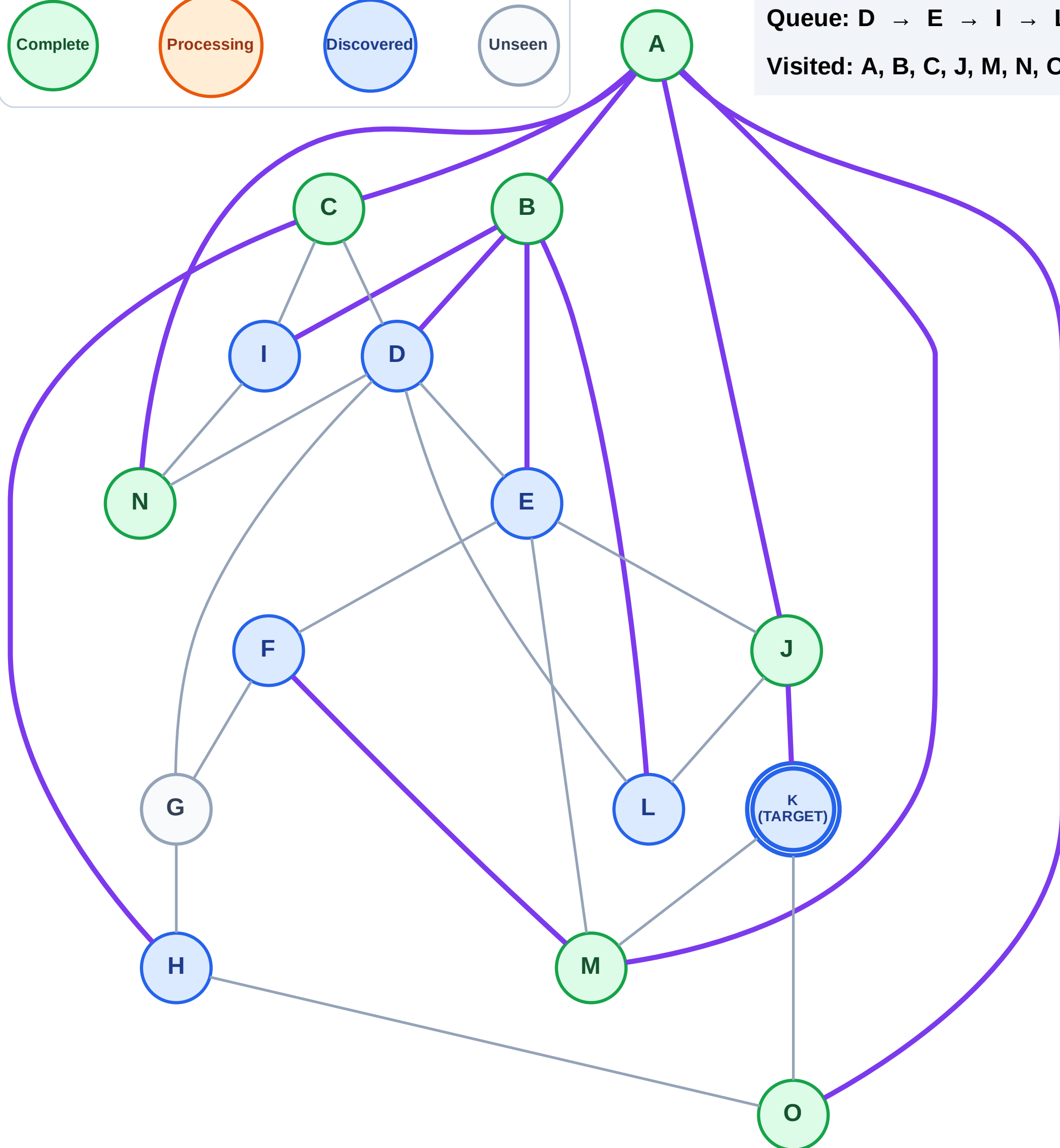
Step 15: No new neighbors from O

Legend



Queue: D → E → I → L → H → K → F

Visited: A, B, C, J, M, N, O



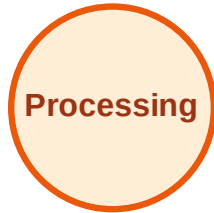
BFS Traversal

Step 16: Process node D

Legend



Complete



Processing



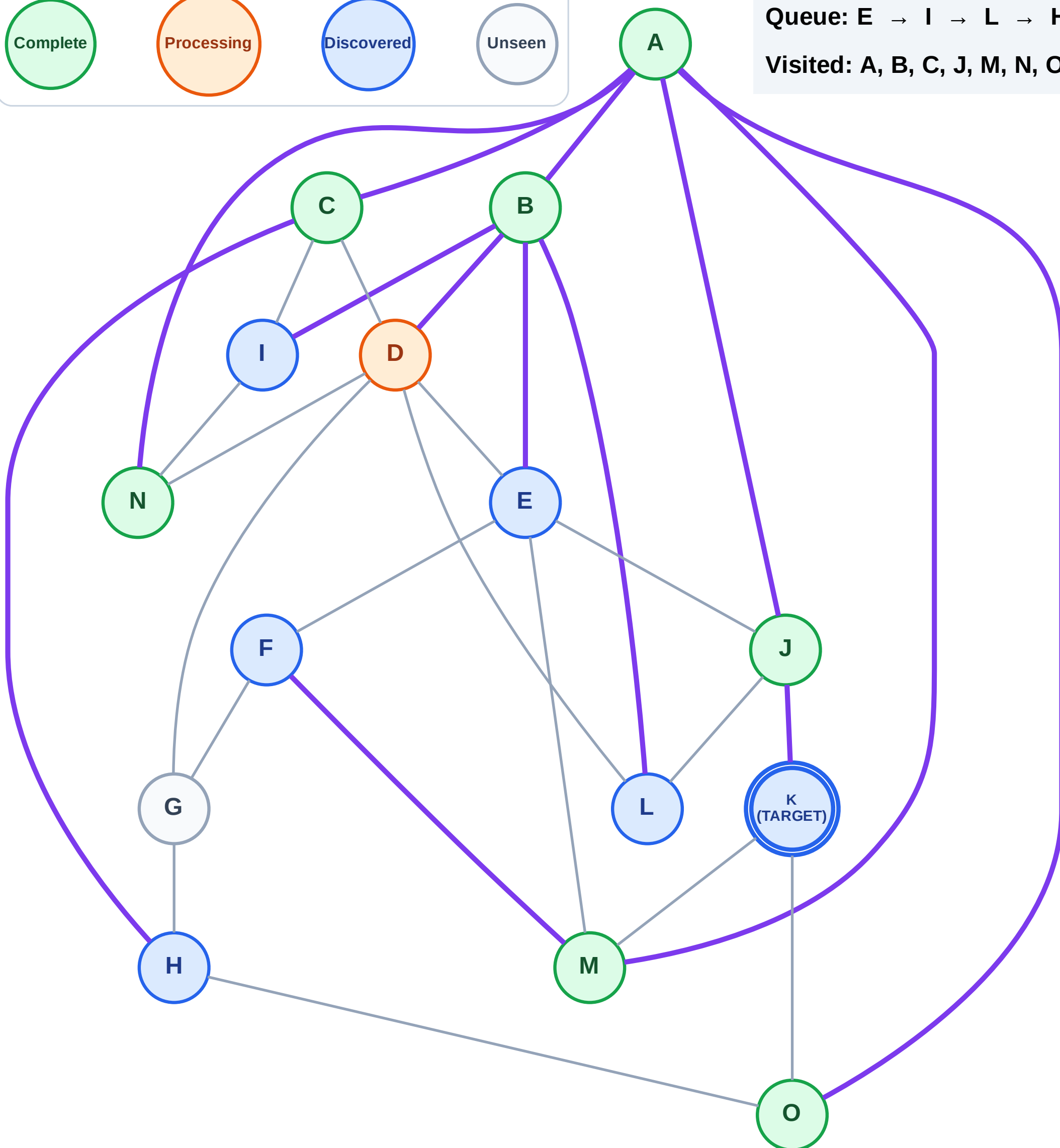
Discovered



Unseen

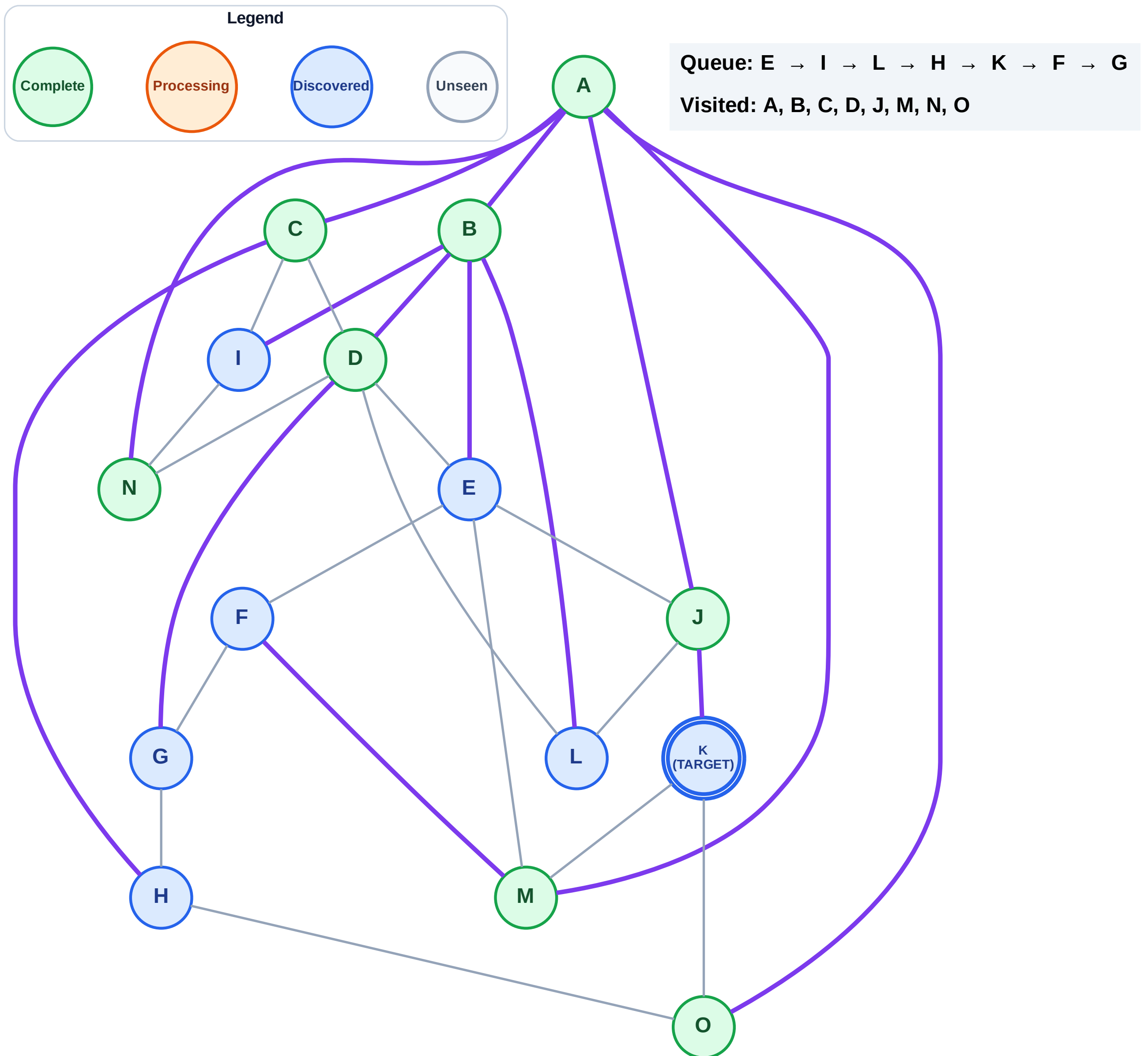
Queue: E → I → L → H → K → F

Visited: A, B, C, J, M, N, O



Step 17: Discover G from D

Step 17: Discover G from D



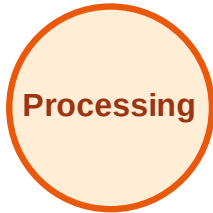
BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



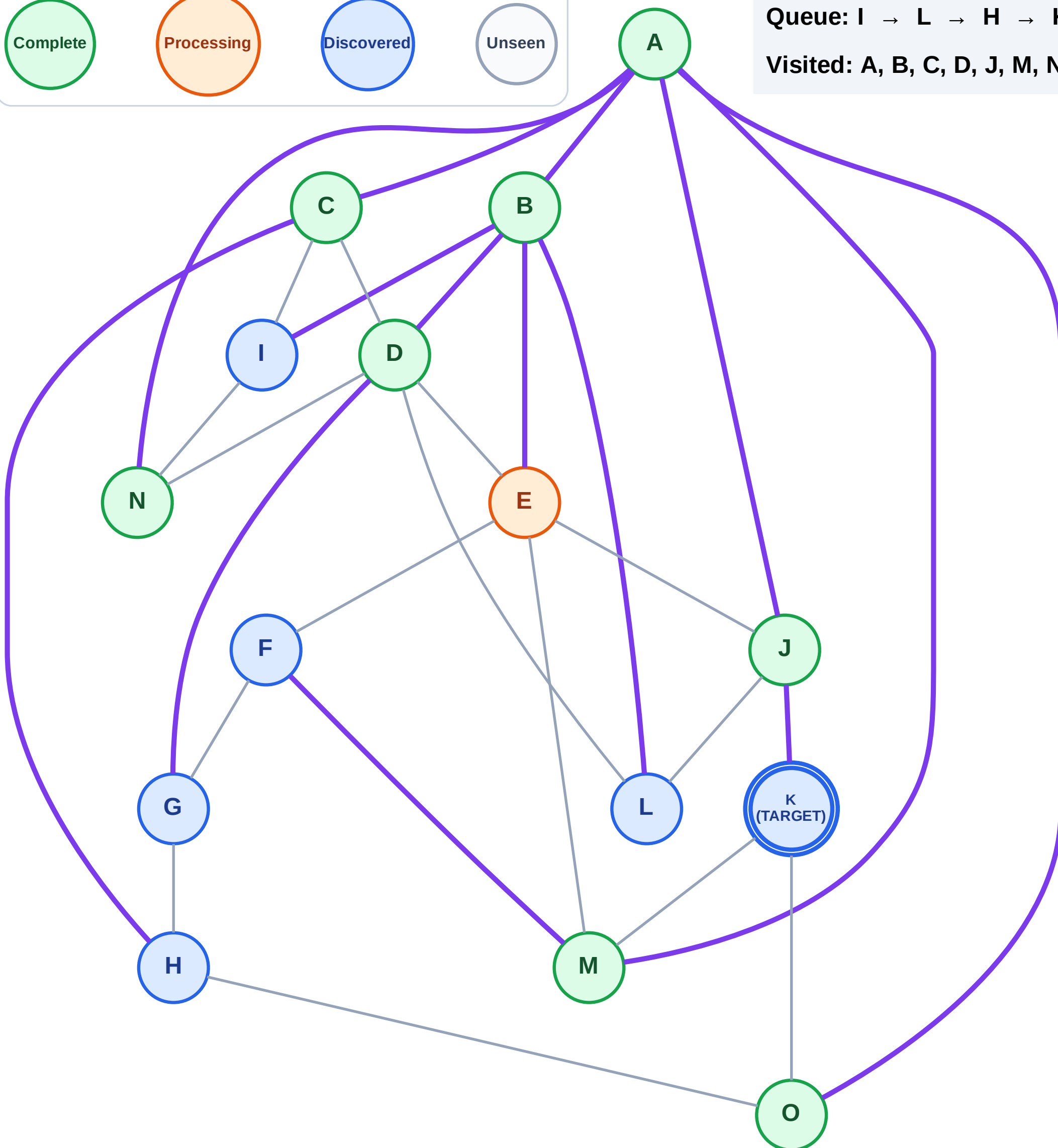
Discovered



Unseen

Queue: I → L → H → K → F → G

Visited: A, B, C, D, J, M, N, O



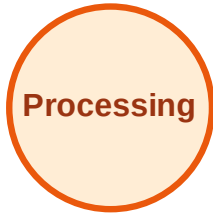
BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



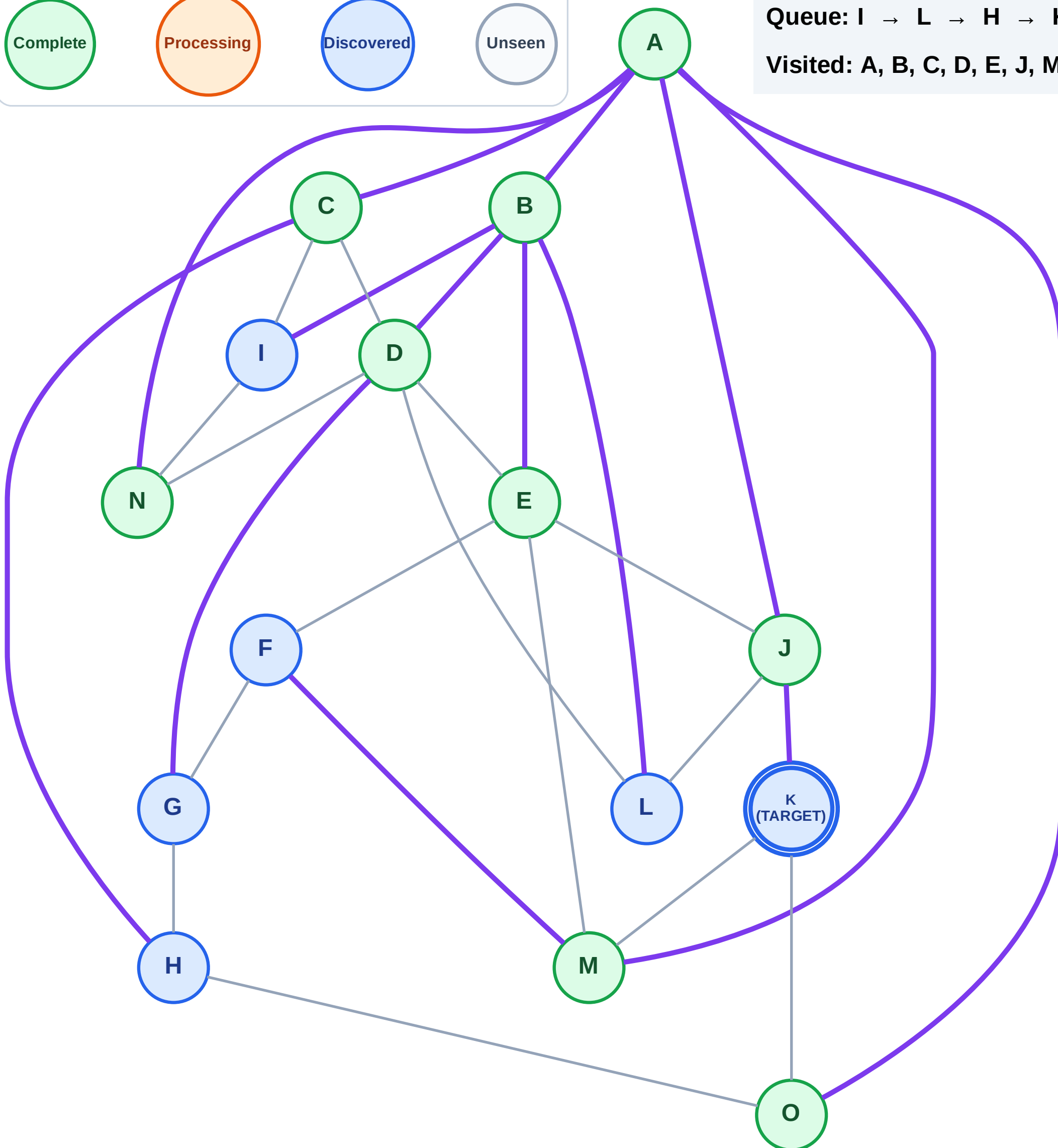
Discovered



Unseen

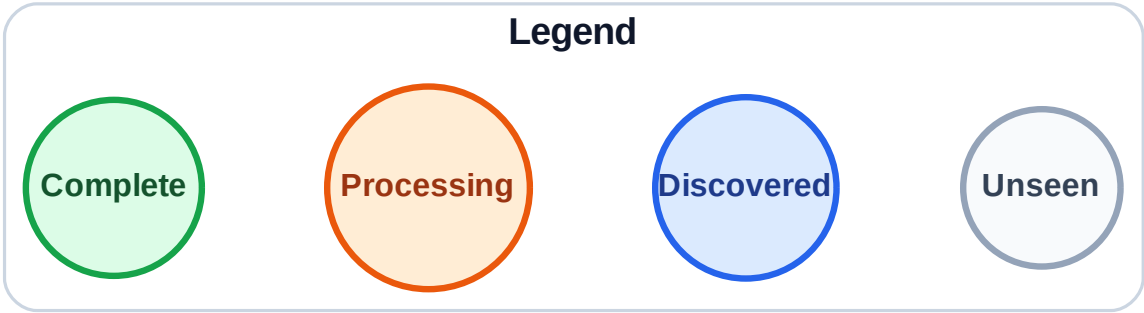
Queue: I → L → H → K → F → G

Visited: A, B, C, D, E, J, M, N, O



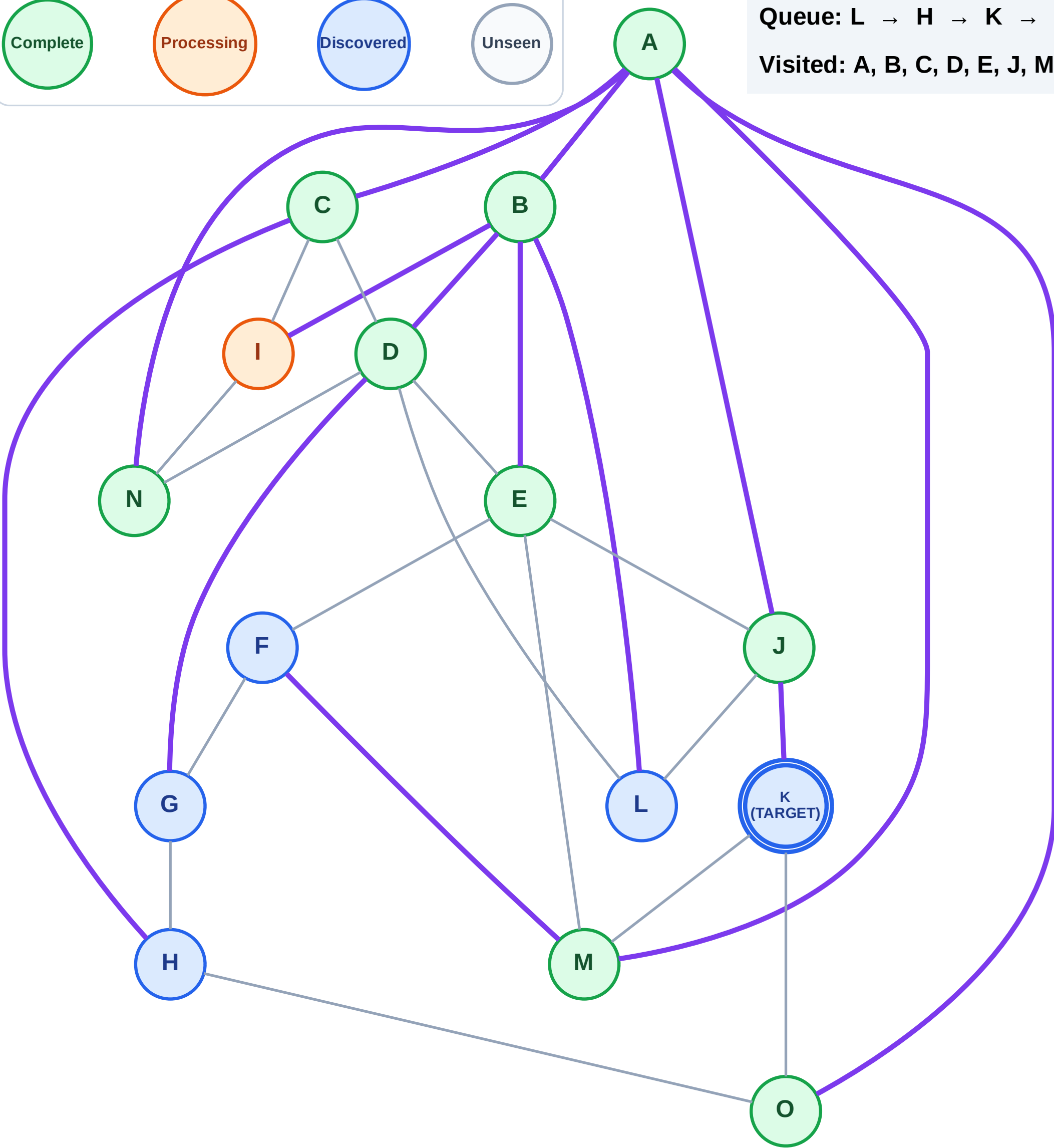
BFS Traversal

Step 20: Process node I



Queue: L → H → K → F → G

Visited: A, B, C, D, E, J, M, N, O



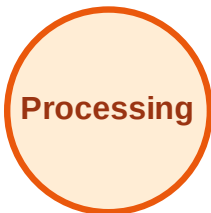
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



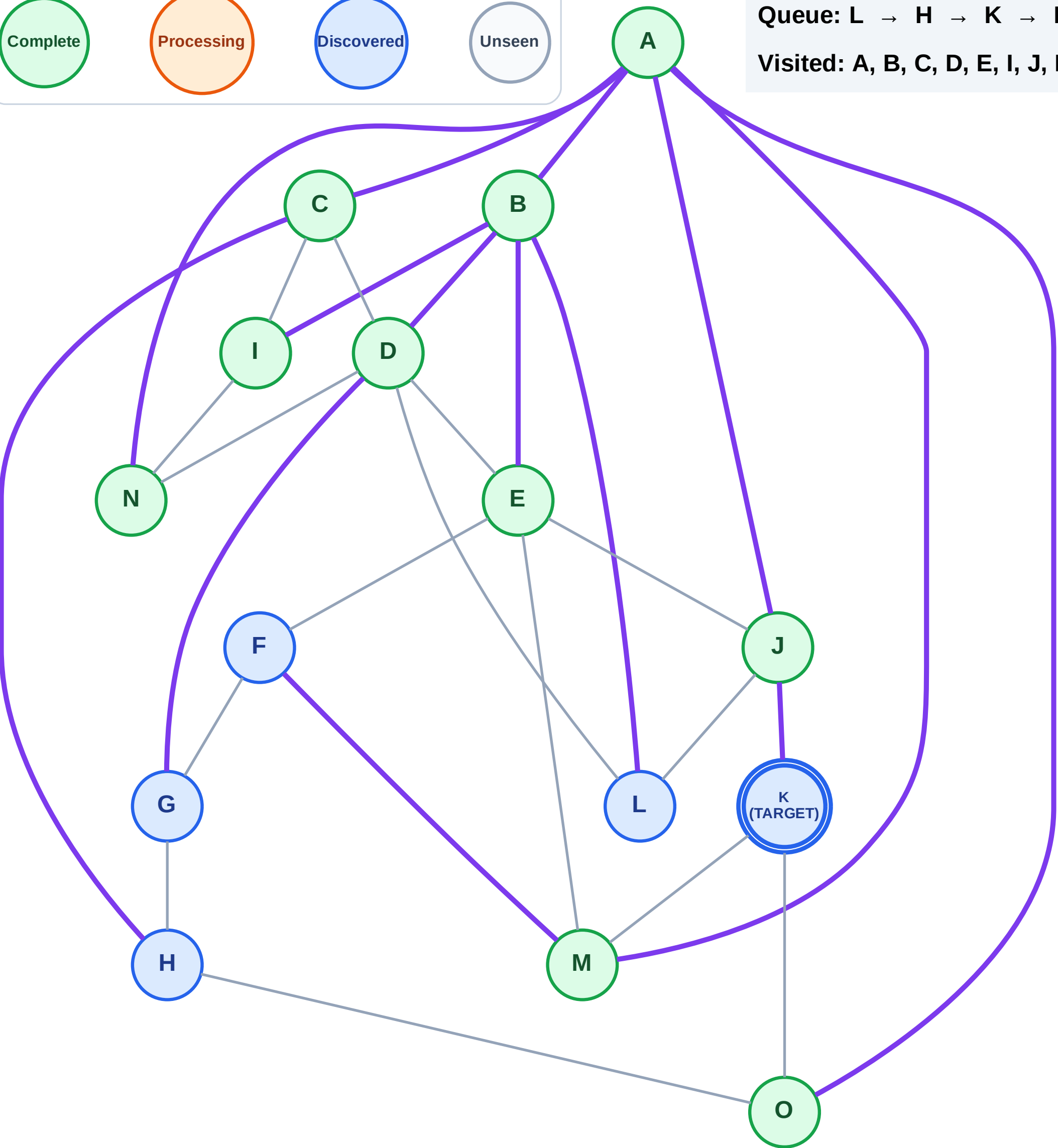
Discovered



Unseen

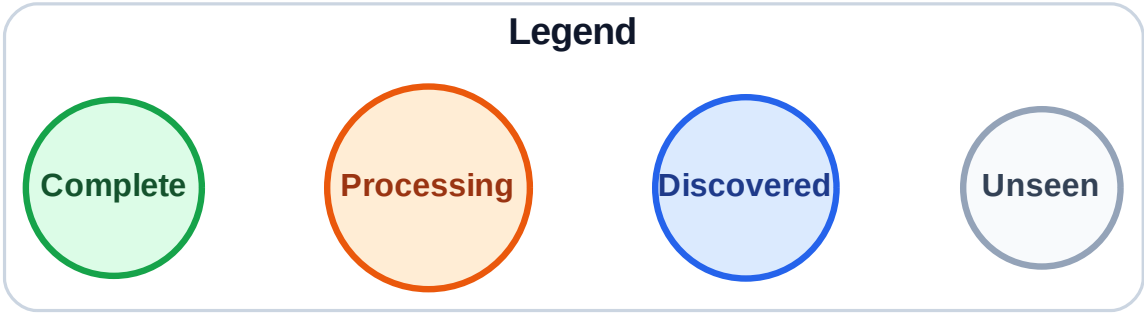
Queue: L → H → K → F → G

Visited: A, B, C, D, E, I, J, M, N, O

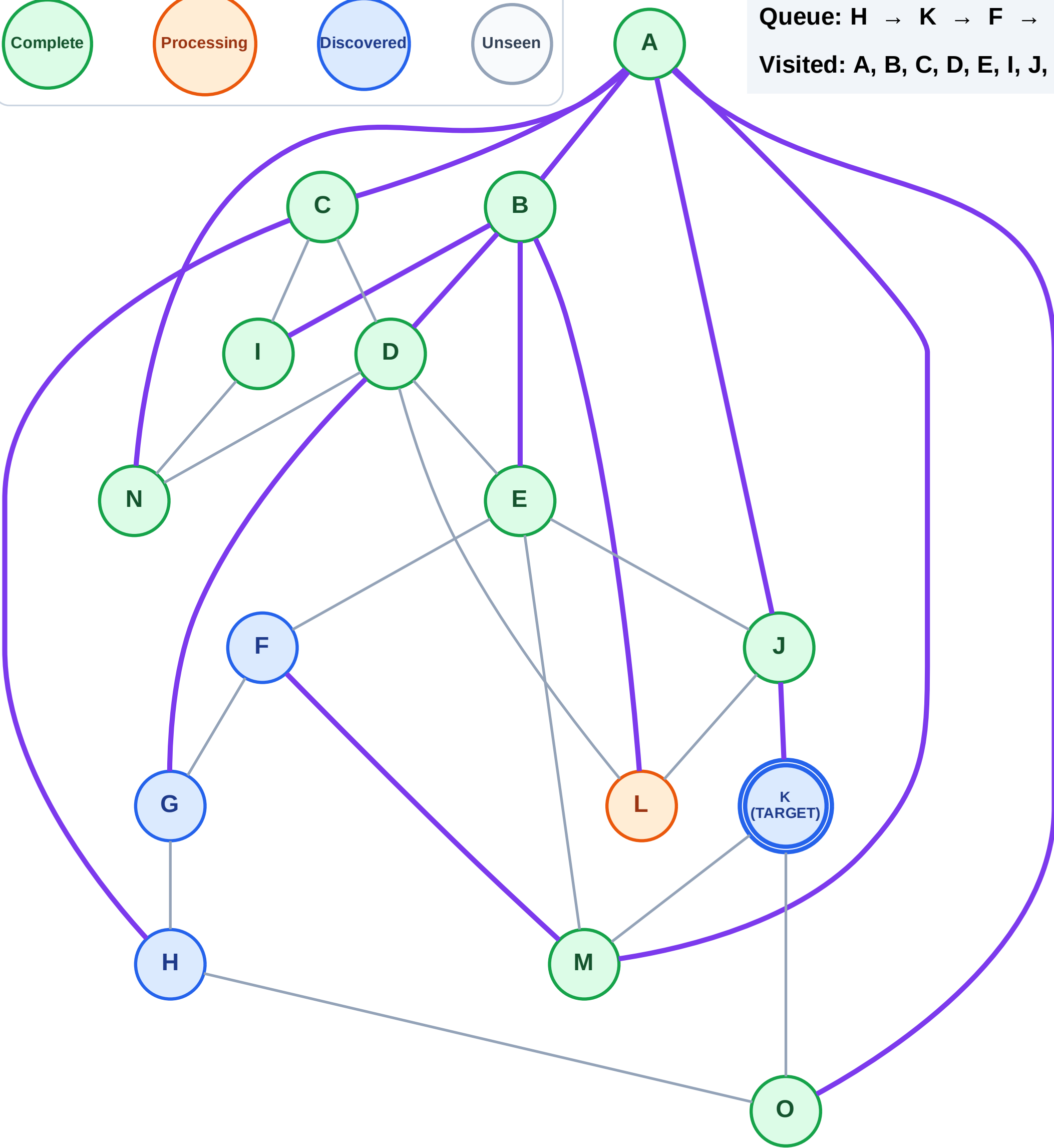


BFS Traversal

Step 22: Process node L

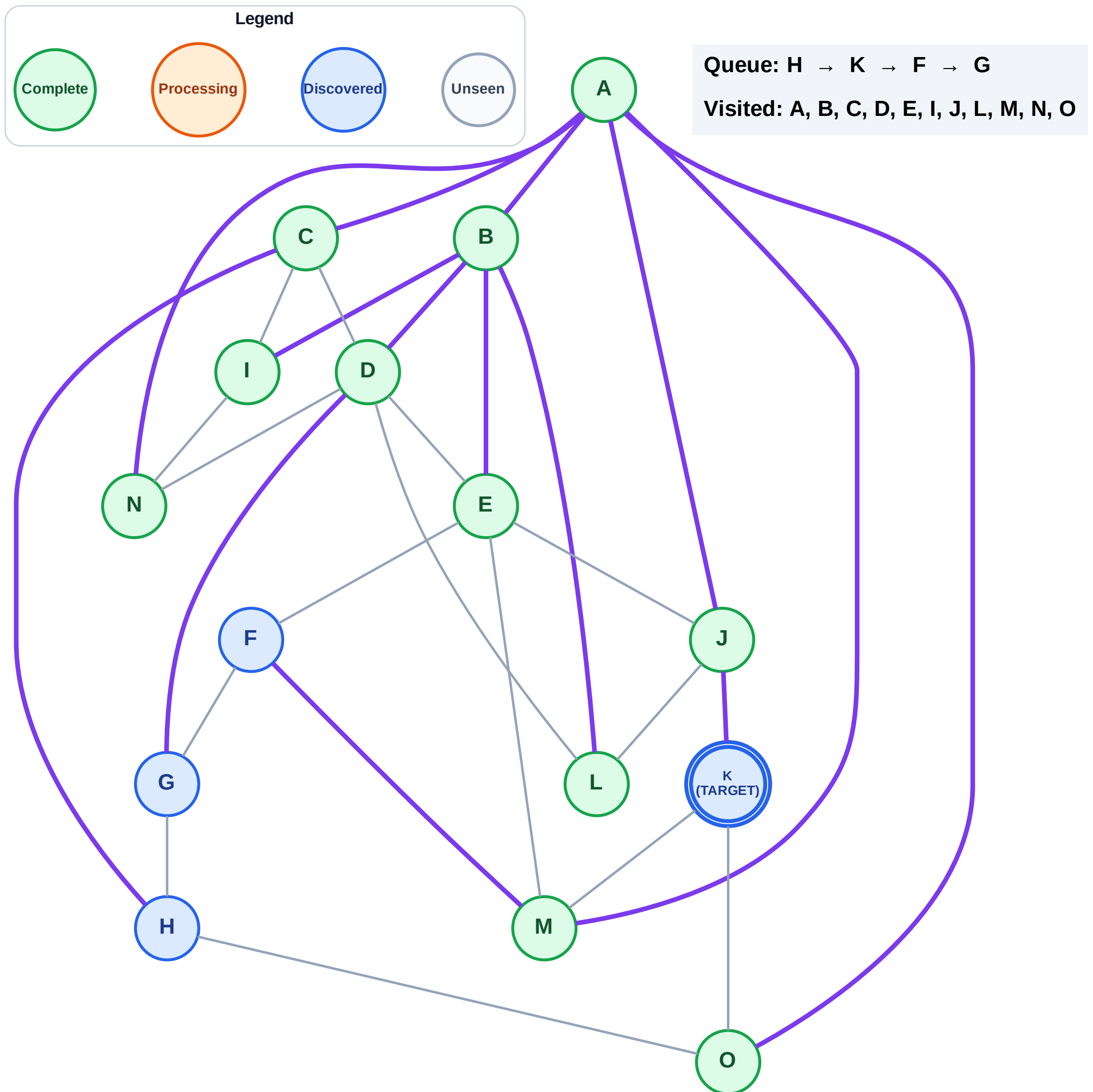


Queue: H → K → F → G
Visited: A, B, C, D, E, I, J, M, N, O

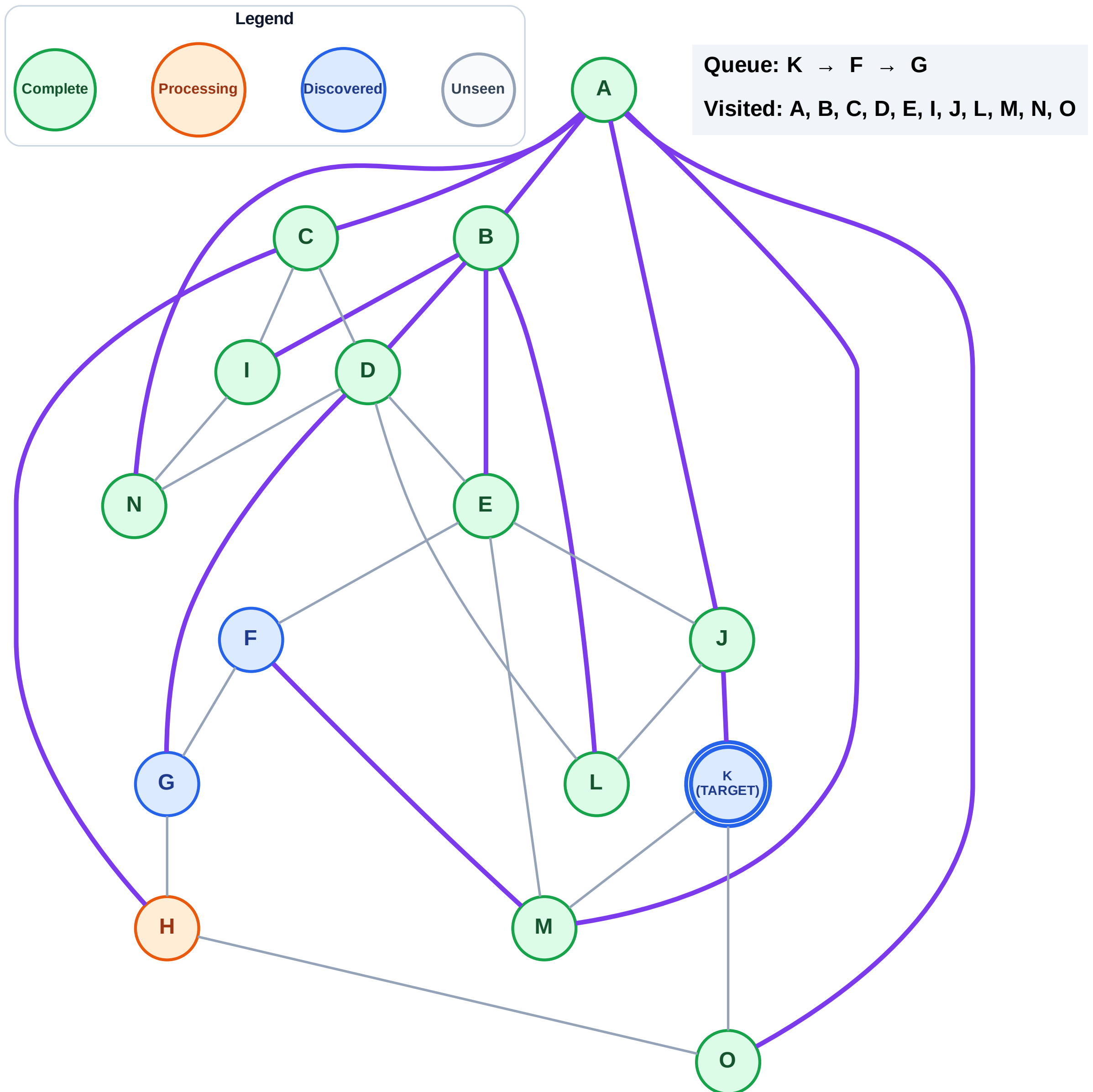


Step 23: No new neighbors from L

Step 23: No new neighbors from L



Step 24: Process node H



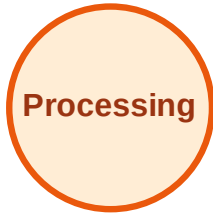
BFS Traversal

Step 25: No new neighbors from H

Legend



Complete



Processing



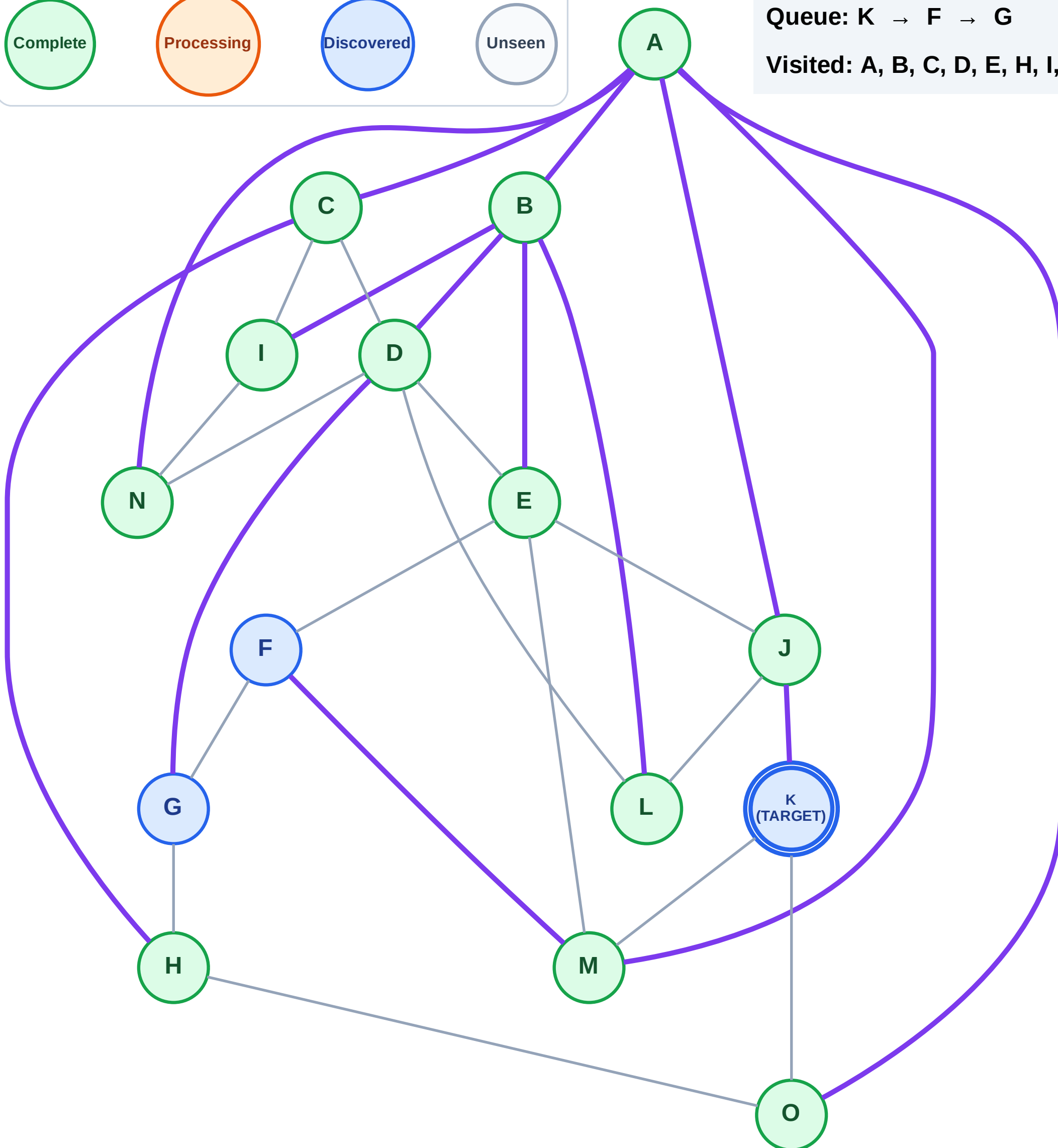
Discovered



Unseen

Queue: K → F → G

Visited: A, B, C, D, E, H, I, J, L, M, N, O



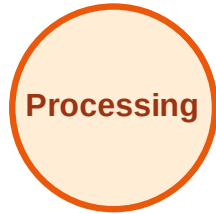
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



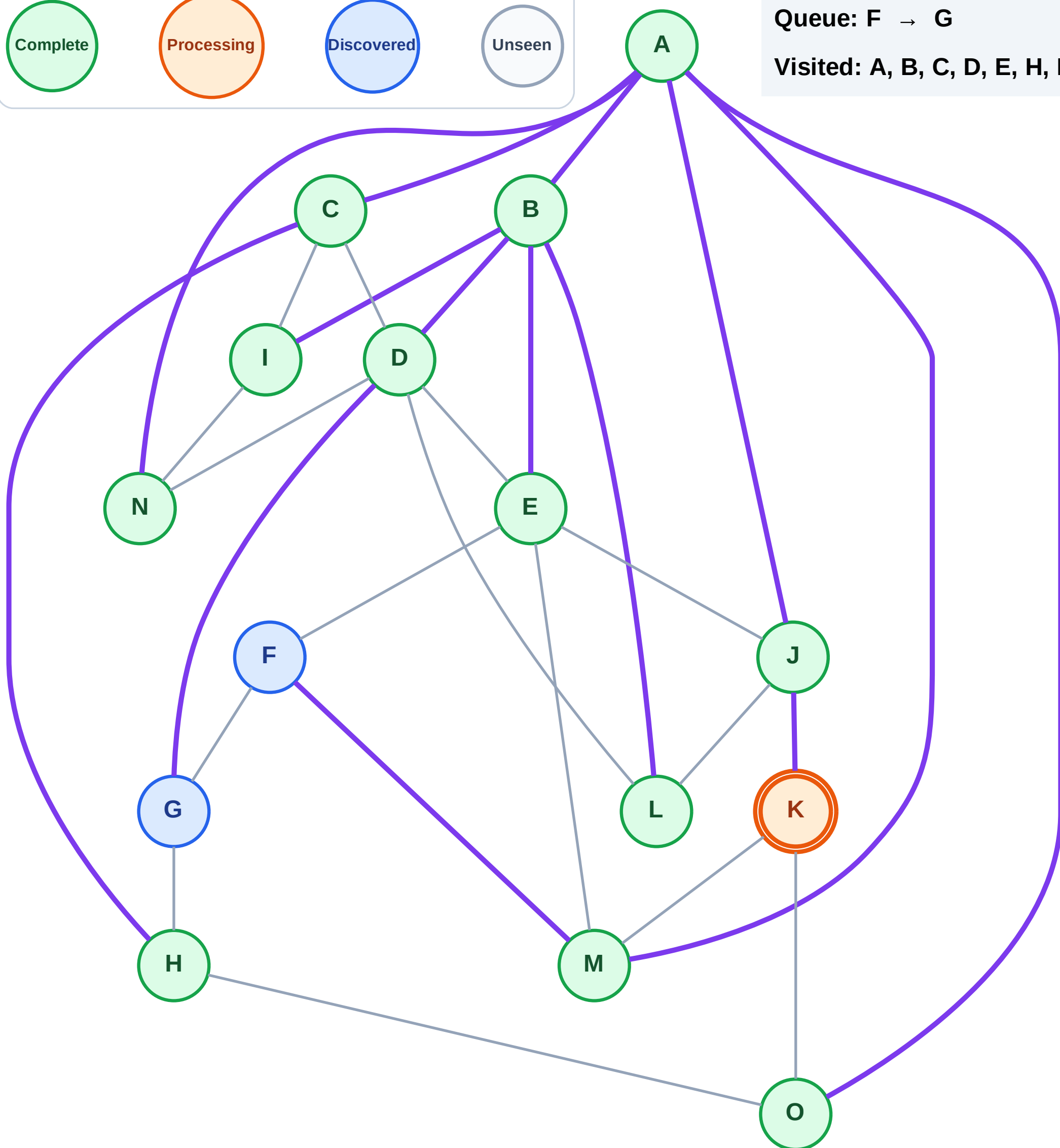
Discovered



Unseen

Queue: F → G

Visited: A, B, C, D, E, H, I, J, L, M, N, O



BFS Traversal

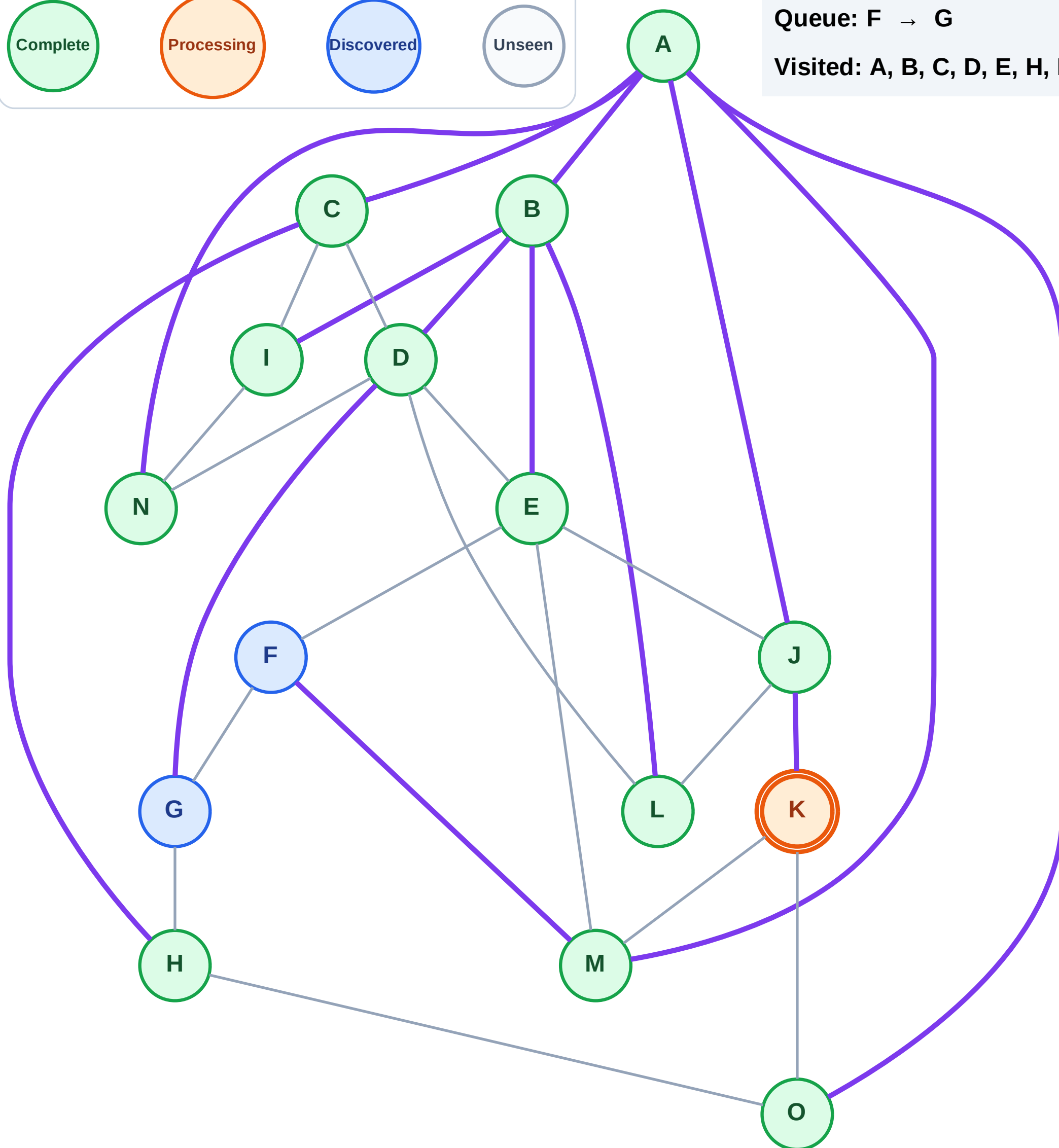
Step 27: TARGET NODE K REACHED! Path (2 hops): A → J → K

Legend



Queue: F → G

Visited: A, B, C, D, E, H, I, J, K, L, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

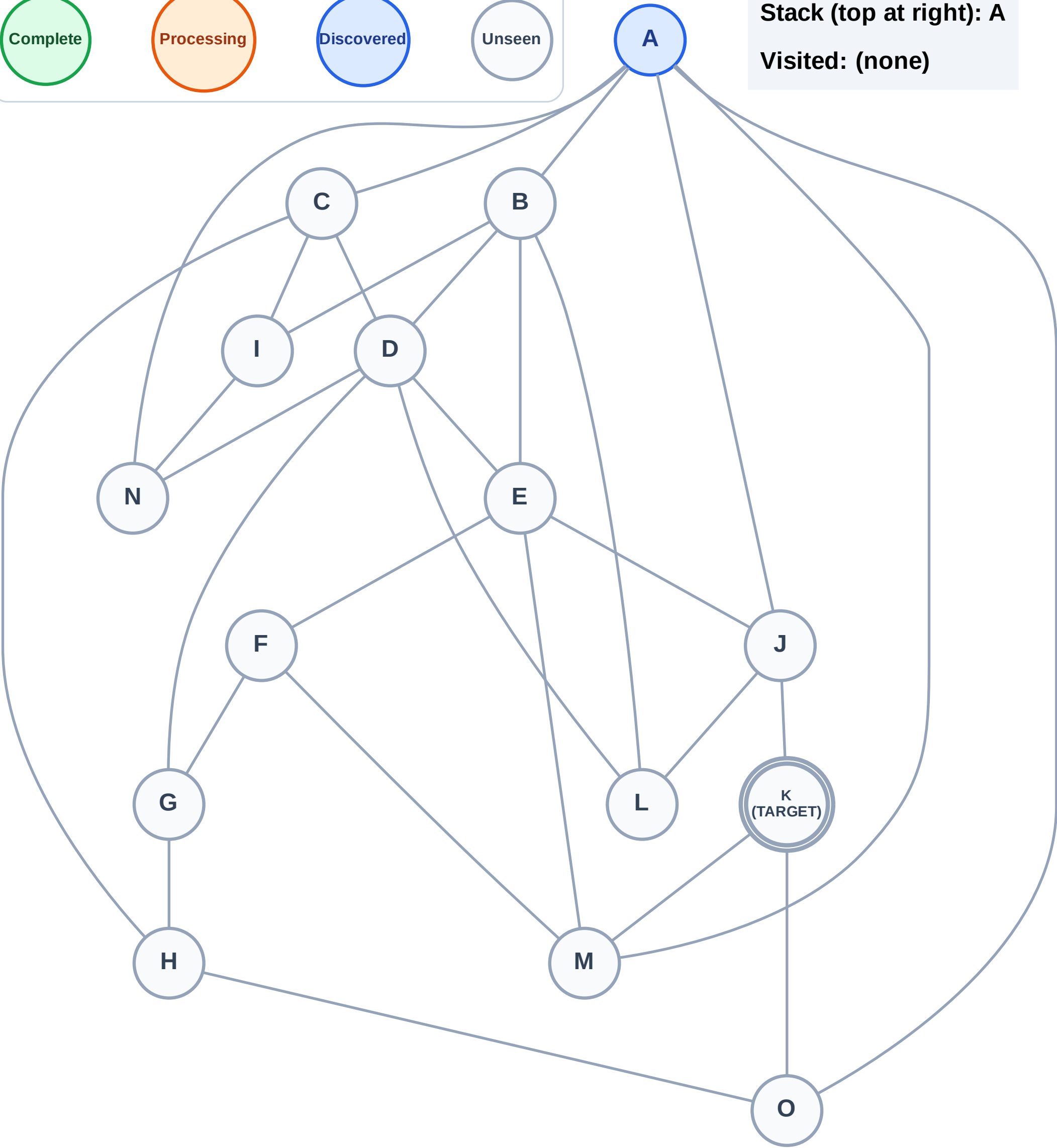
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

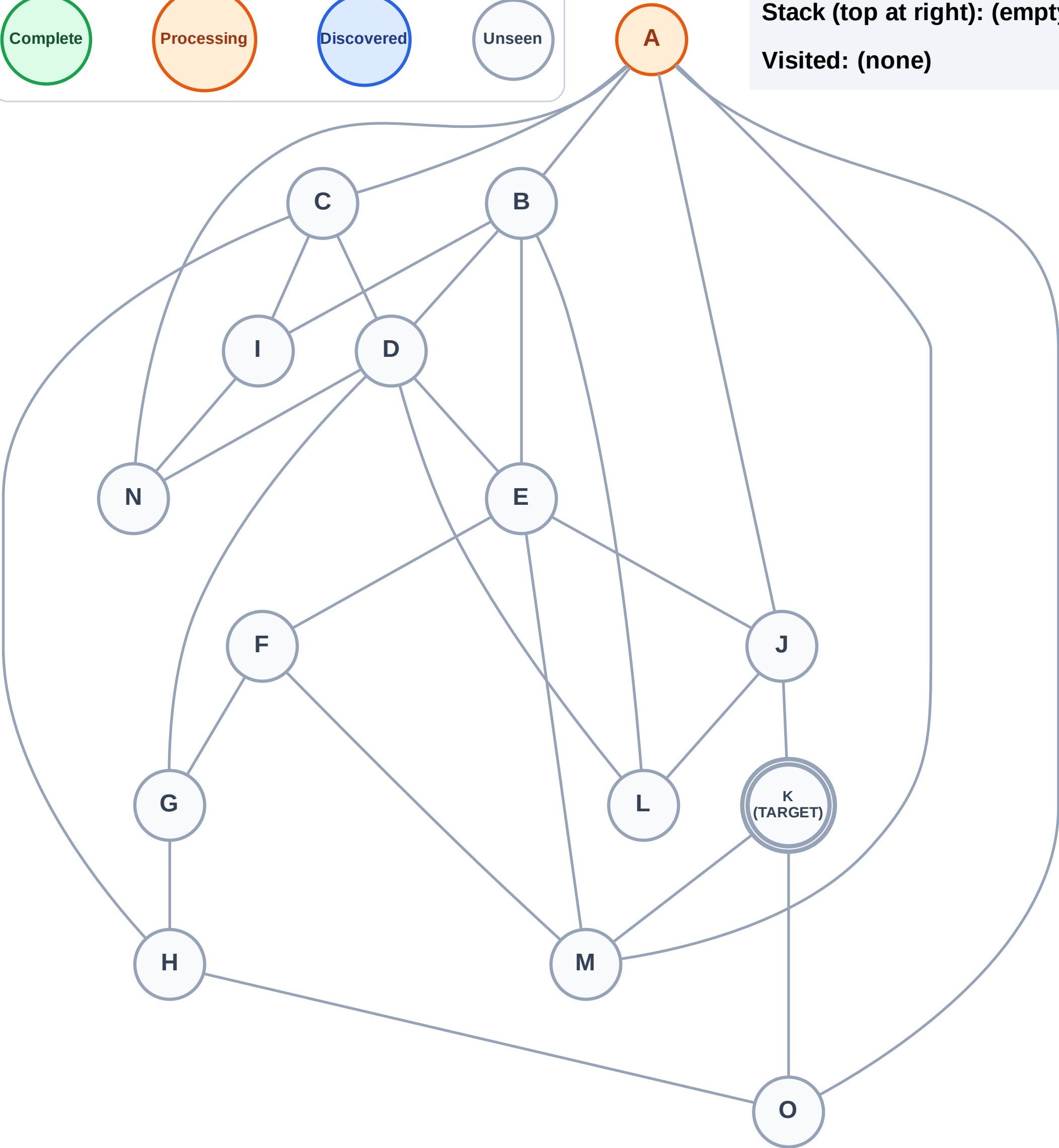
Complete

Processing

Discovered

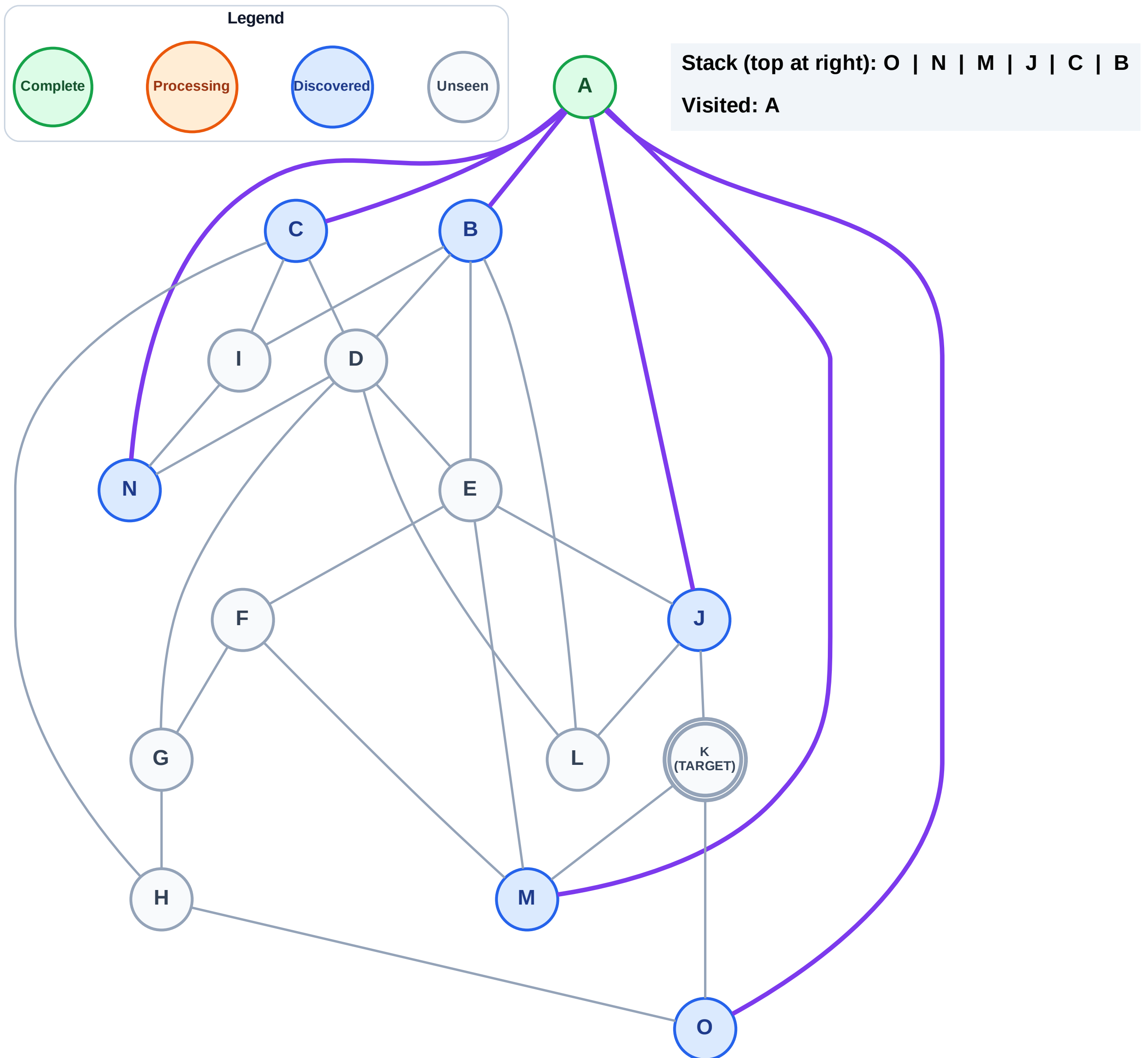
Unseen

Stack (top at right): (empty)
Visited: (none)



Step 3: Discover O, N, M, J, C, B from A

Step 3: Discover O, N, M, J, C, B from A



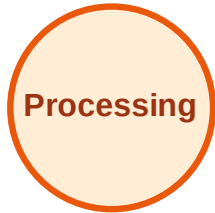
DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



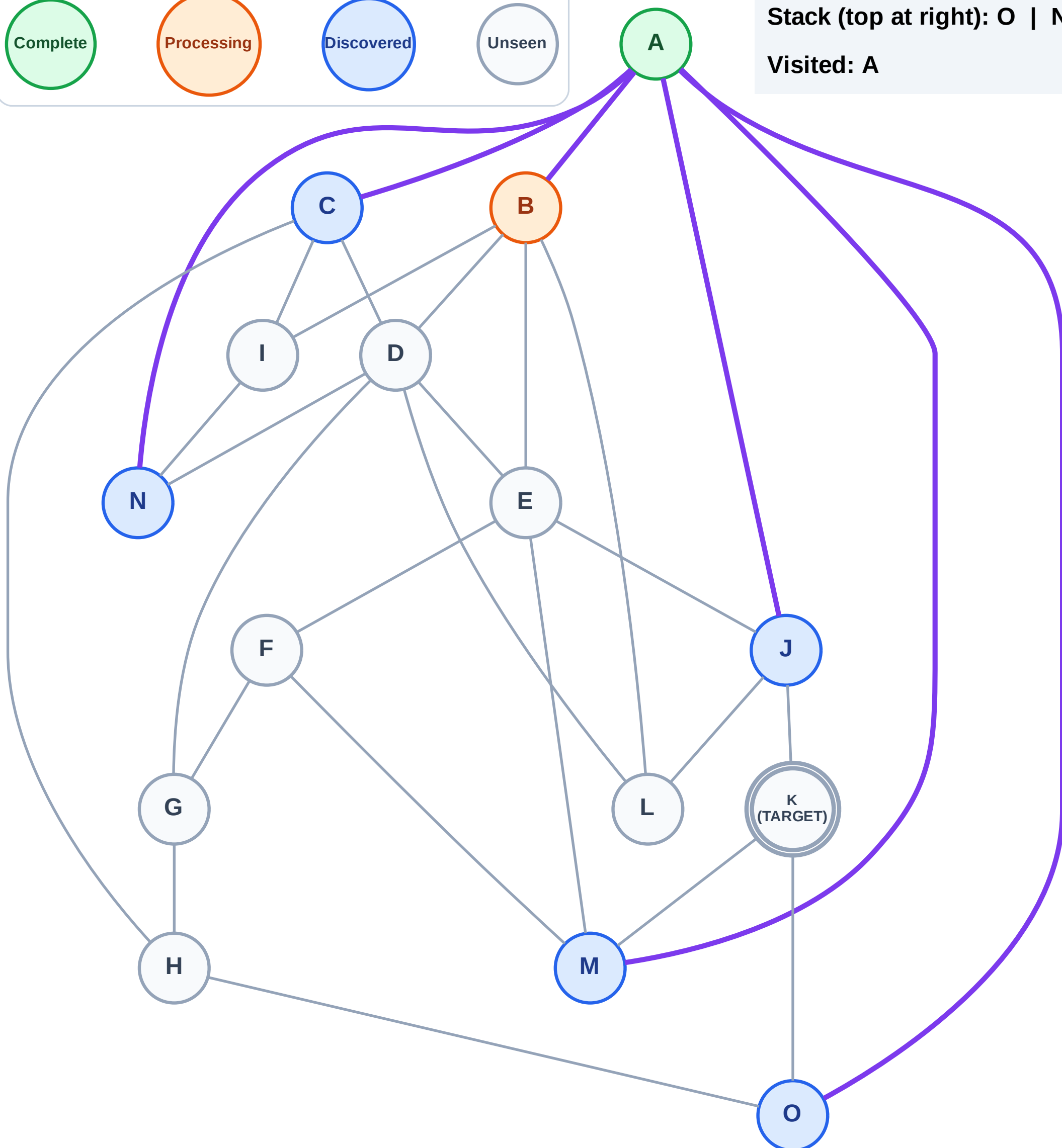
Discovered



Unseen

Stack (top at right): O | N | M | J | C

Visited: A



DFS Traversal

Step 5: Discover L, I, E, D from B

Legend

Complete

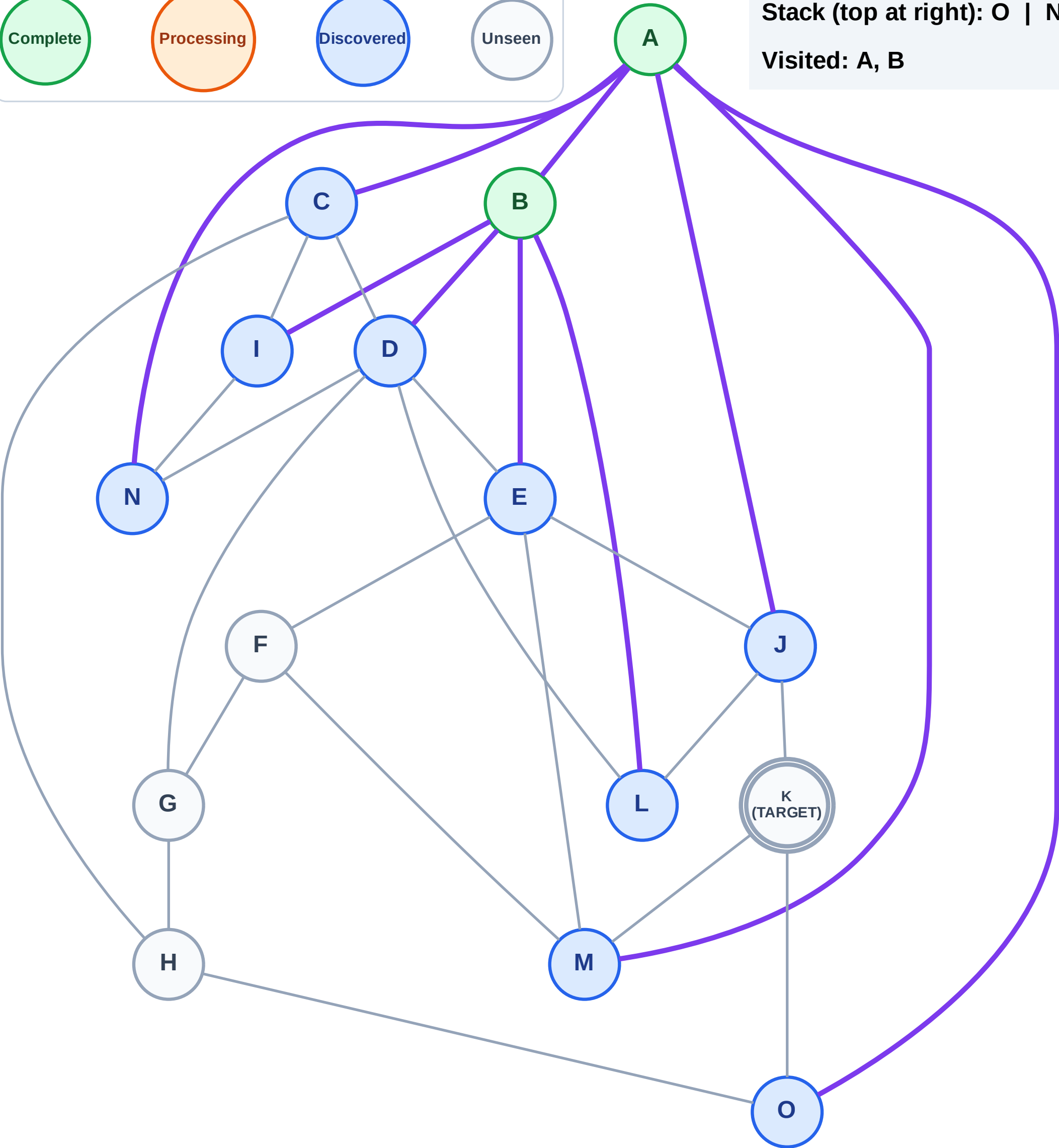
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C | L | I | E | D

Visited: A, B



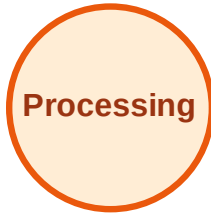
DFS Traversal

Step 6: Process node D

Legend



Complete



Processing



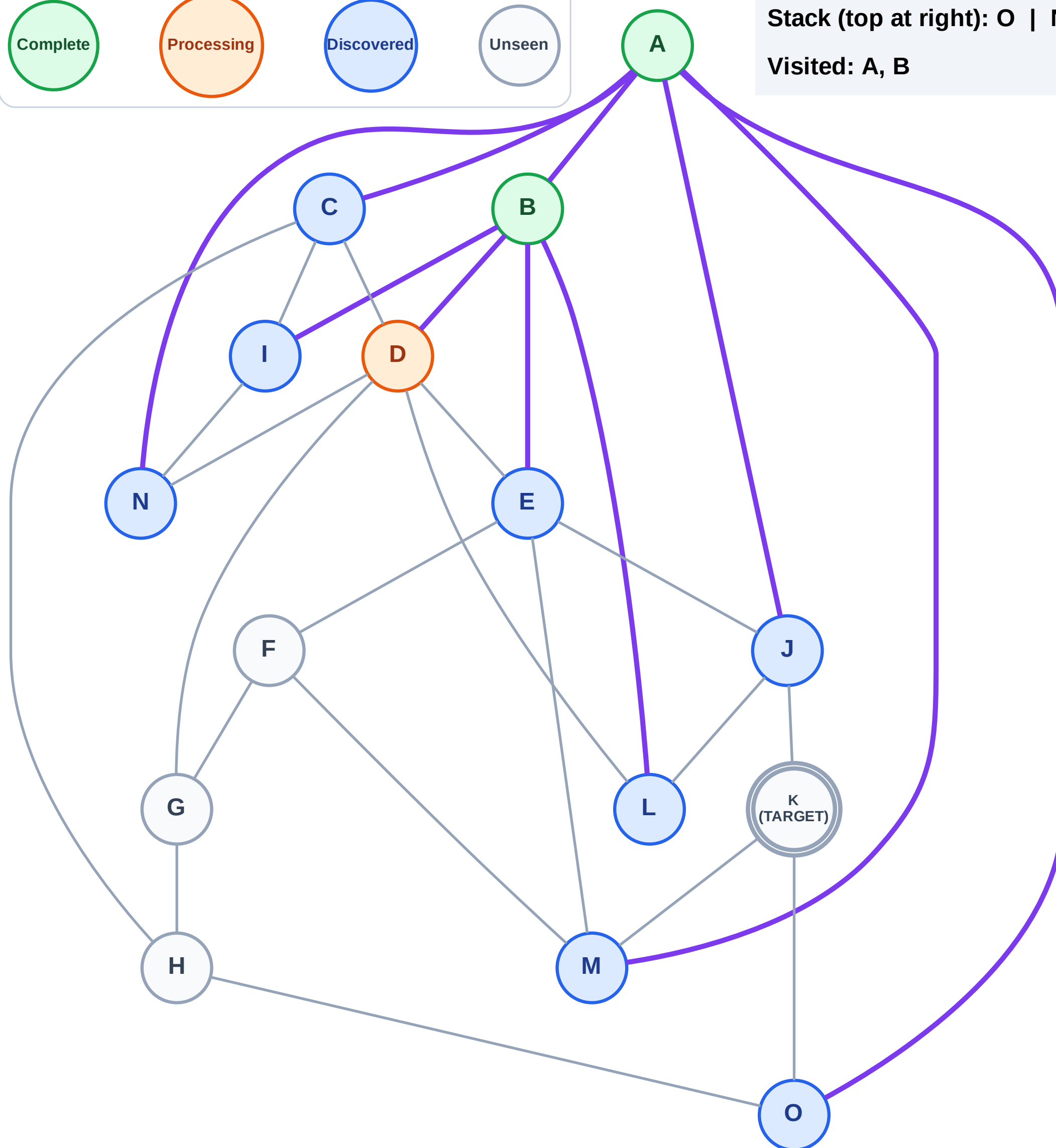
Discovered



Unseen

Stack (top at right): O | N | M | J | C | L | I | E

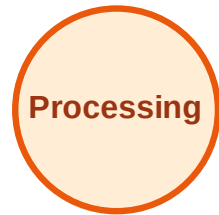
Visited: A, B



DFS Traversal

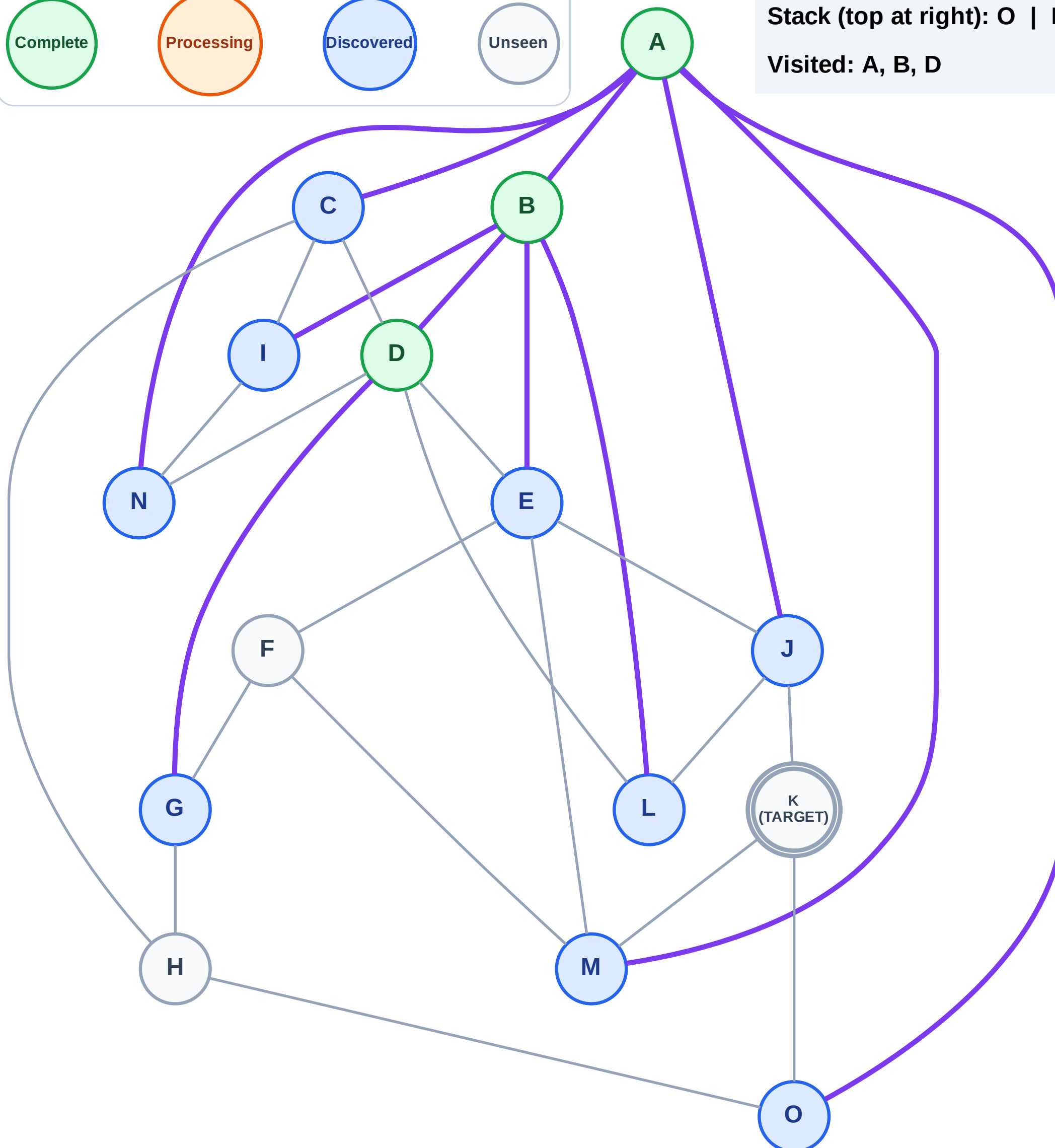
Step 7: Discover G from D

Legend



Stack (top at right): O | N | M | J | C | L | I | E | G

Visited: A, B, D



DFS Traversal

Step 8: Process node G

Legend

Complete

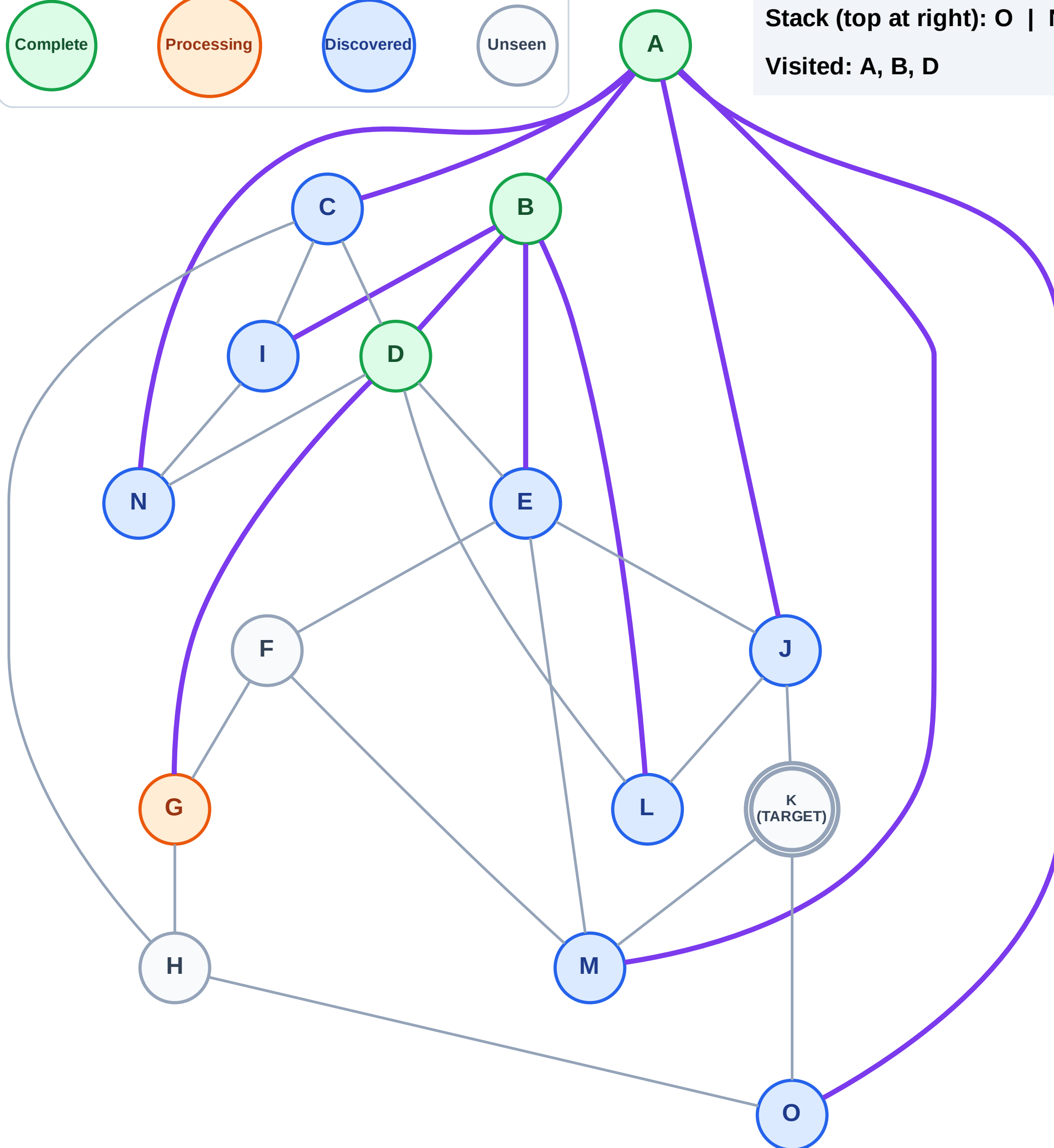
Processing

Discovered

Unseen

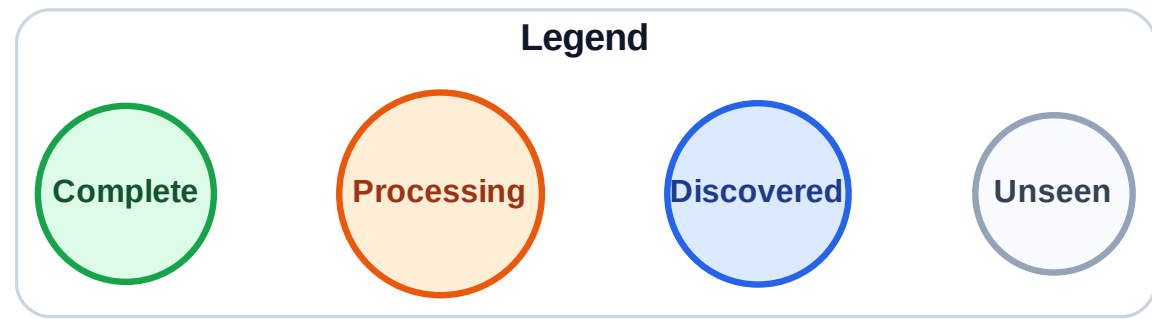
Stack (top at right): O | N | M | J | C | L | I | E

Visited: A, B, D



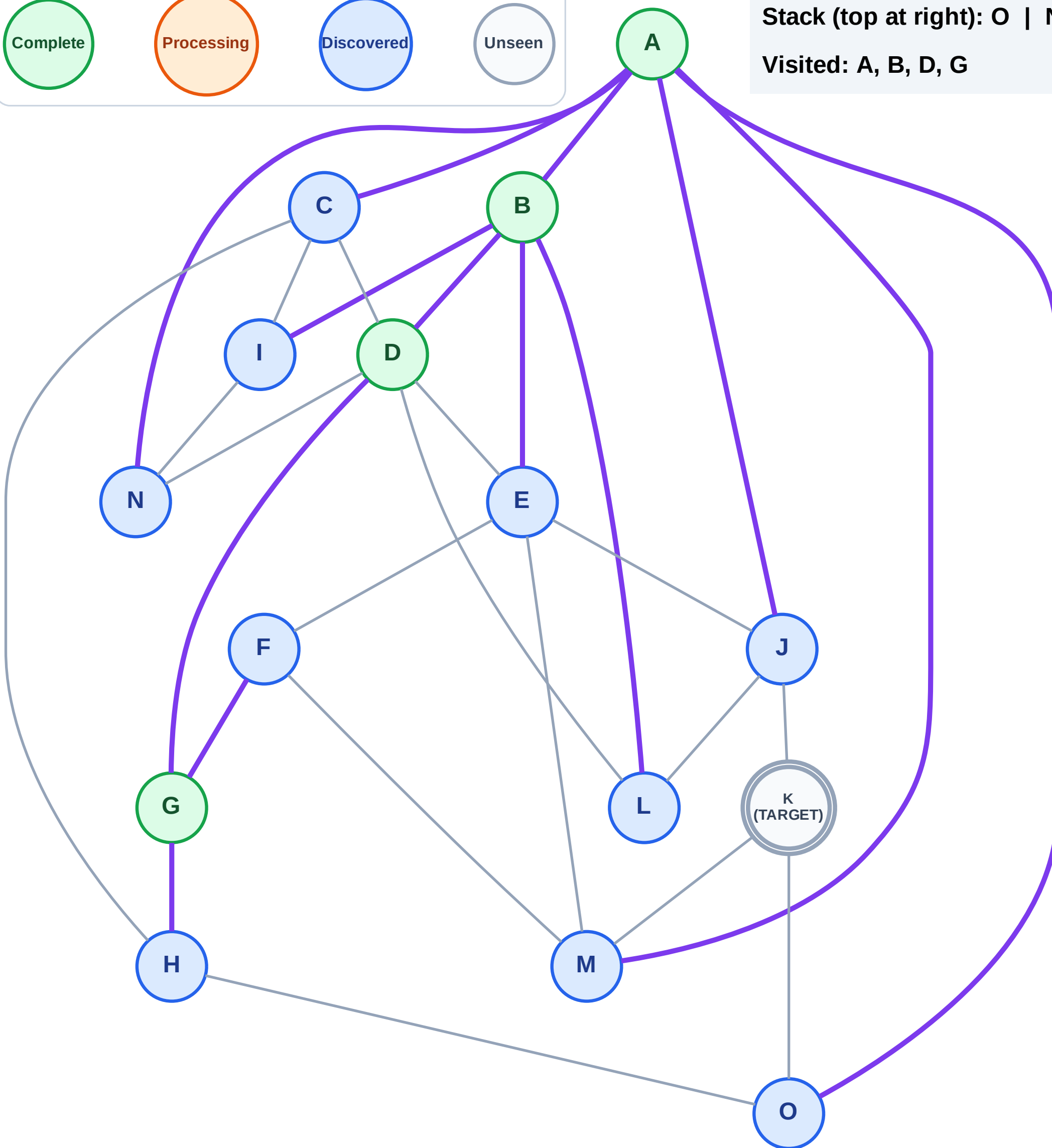
DFS Traversal

Step 9: Discover H, F from G



Stack (top at right): O | N | M | J | C | L | I | E | H | F

Visited: A, B, D, G

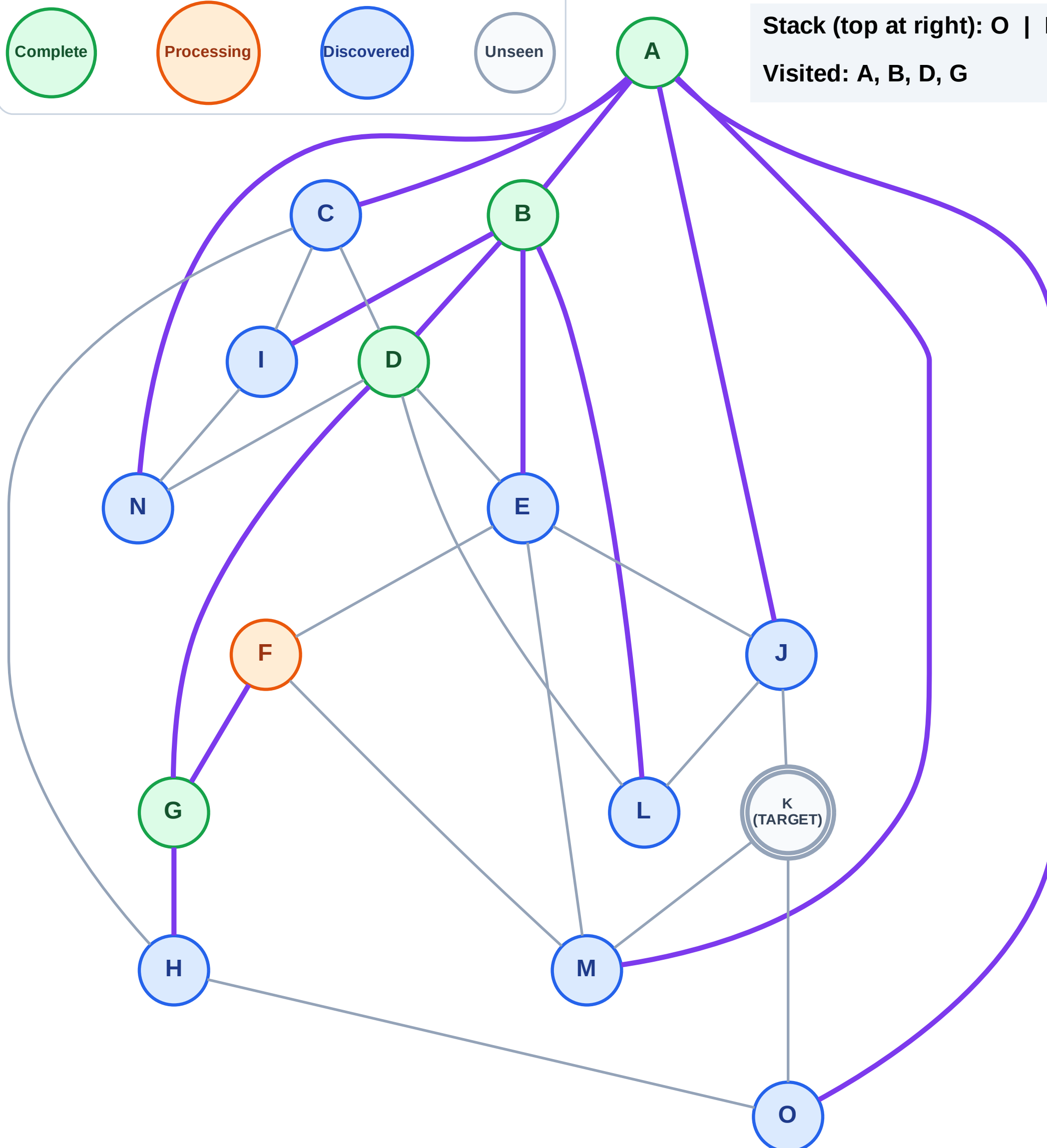


Step 10: Process node F

Stack (top at right): O | N | M | J | C | L | I | E | H

Legend

Unseen



Step 11: No new neighbors from F

Stack (top at right): O | N | M | J | C | L | I | E | H

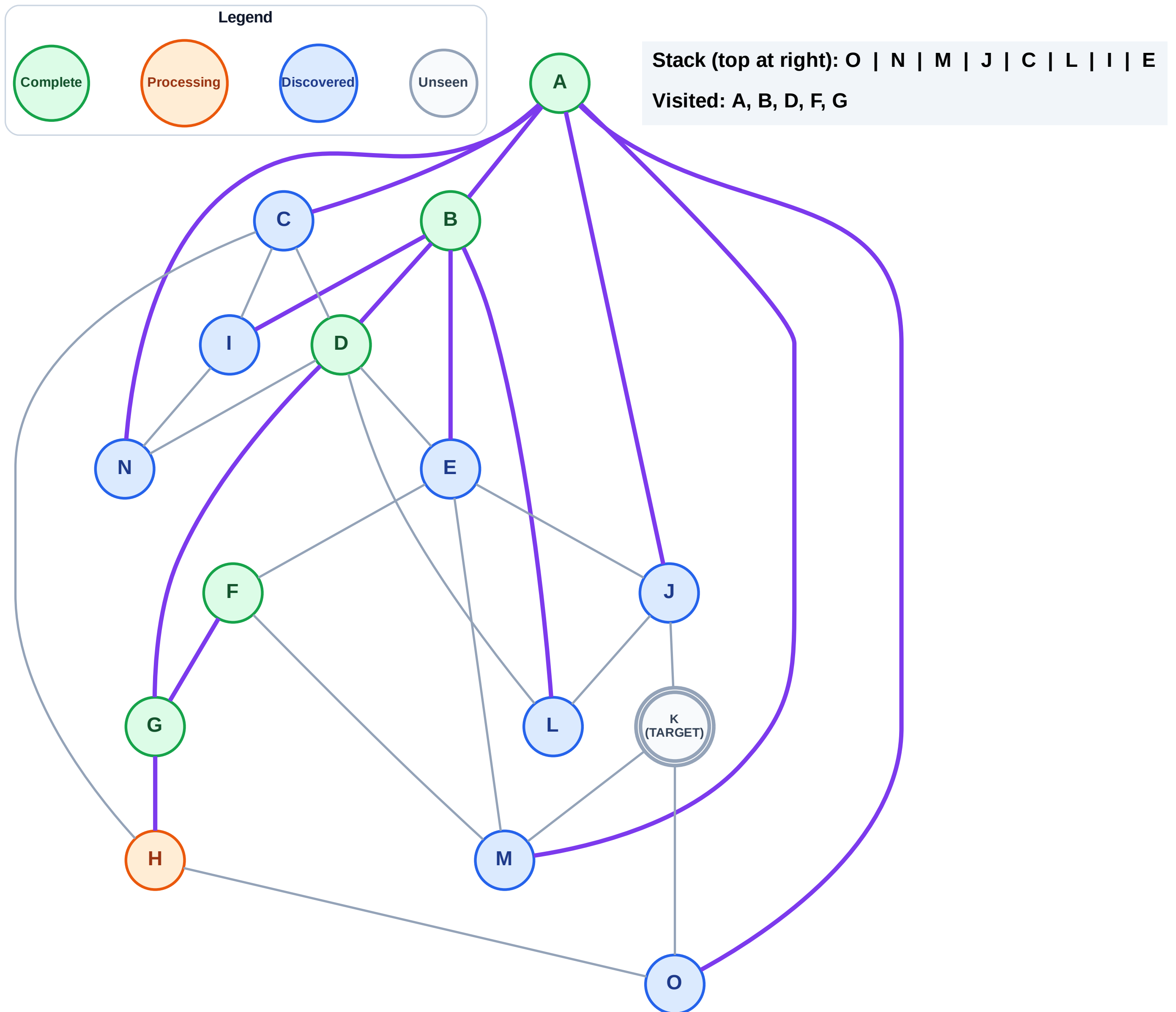
Legend

Unseen



Step 12: Process node H

Step 12: Process node H



DFS Traversal

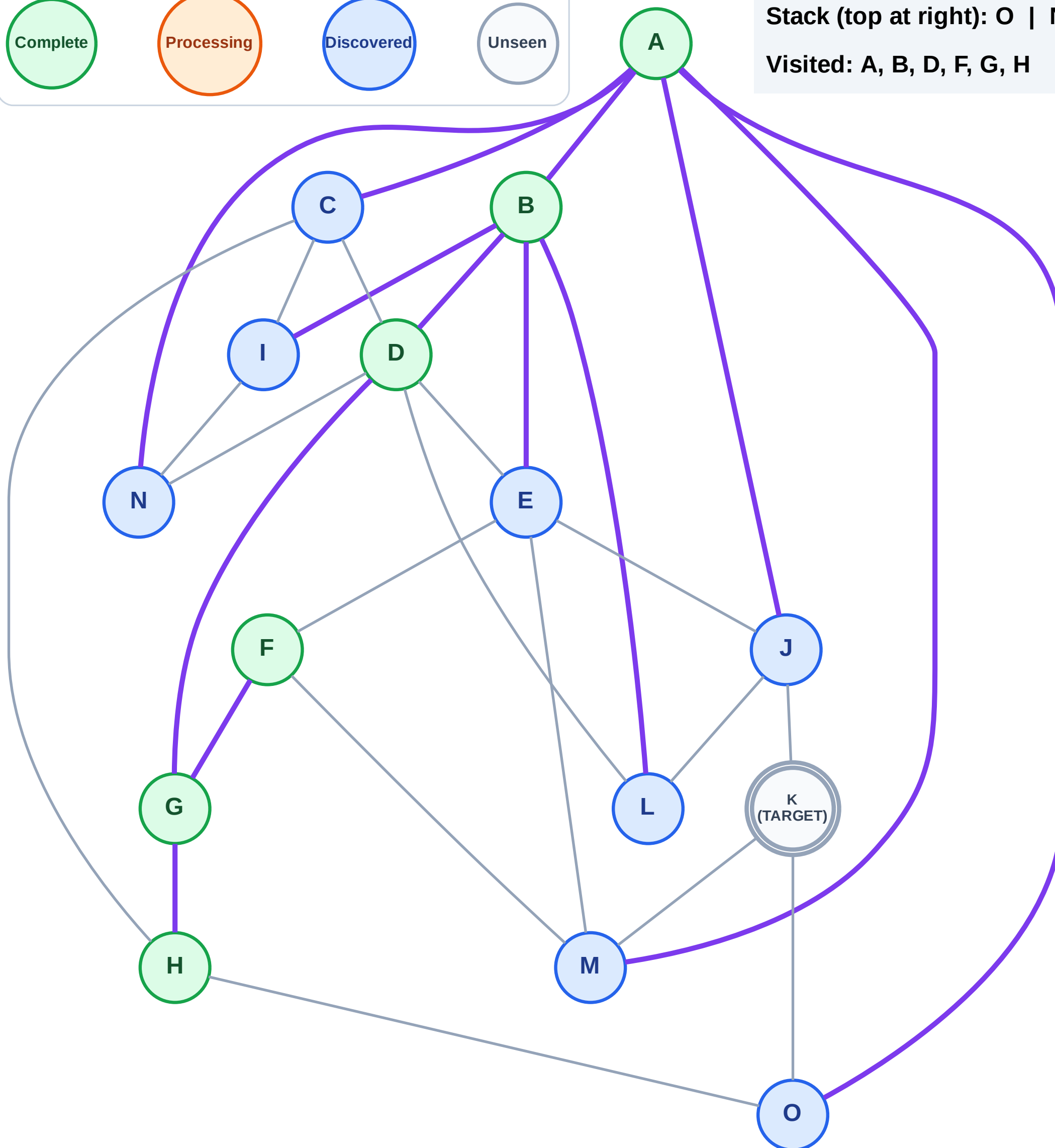
Step 13: No new neighbors from H

Legend



Stack (top at right): O | N | M | J | C | L | I | E

Visited: A, B, D, F, G, H



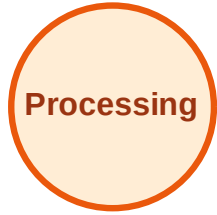
DFS Traversal

Step 14: Process node E

Legend



Complete



Processing



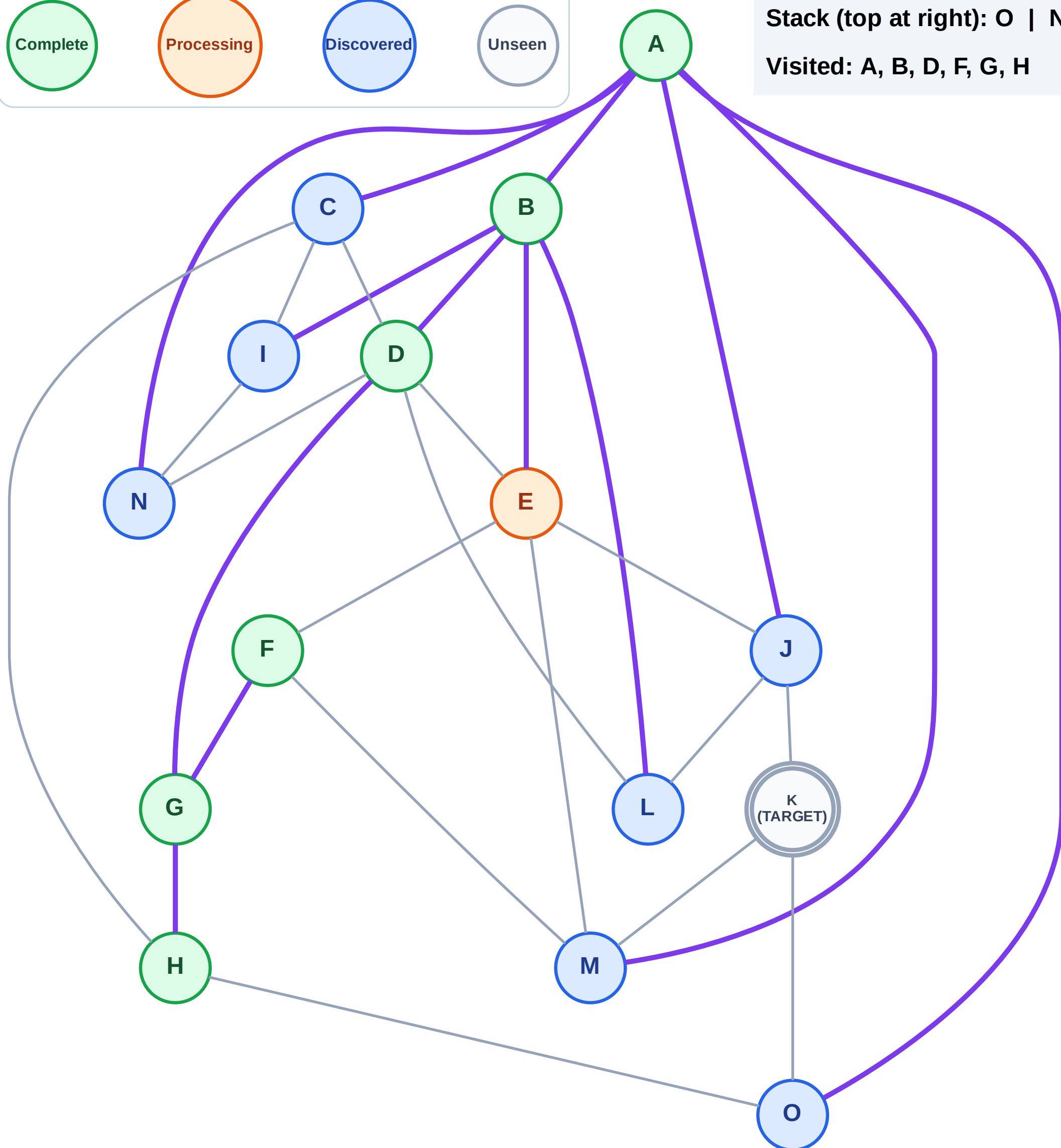
Discovered



Unseen

Stack (top at right): O | N | M | J | C | L | I

Visited: A, B, D, F, G, H



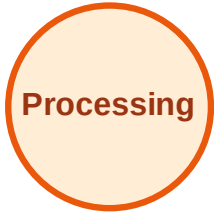
DFS Traversal

Step 15: No new neighbors from E

Legend



Complete



Processing



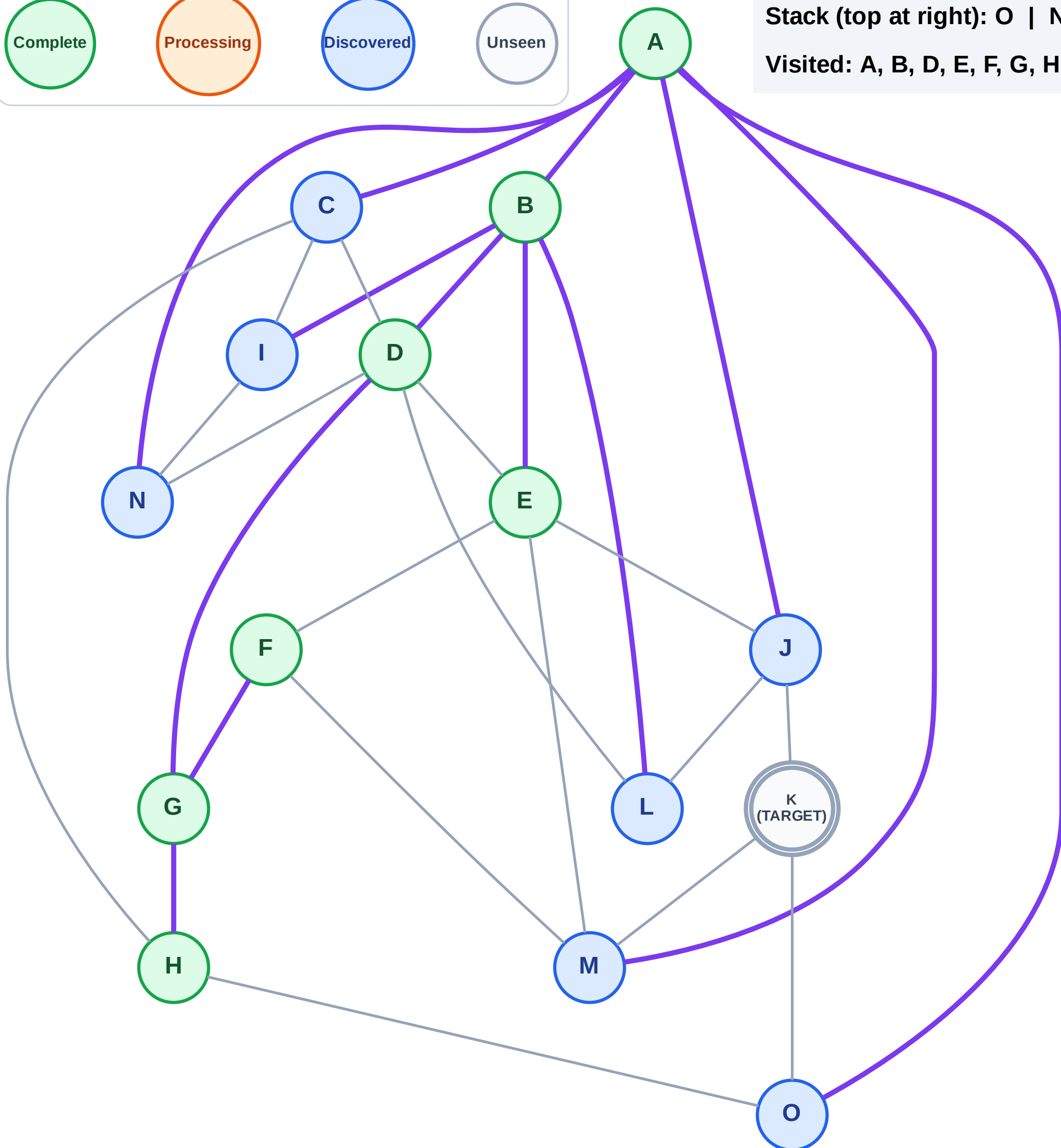
Discovered



Unseen

Stack (top at right): O | N | M | J | C | L | I

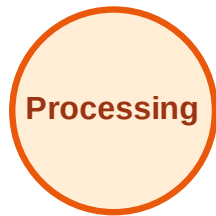
Visited: A, B, D, E, F, G, H



DFS Traversal

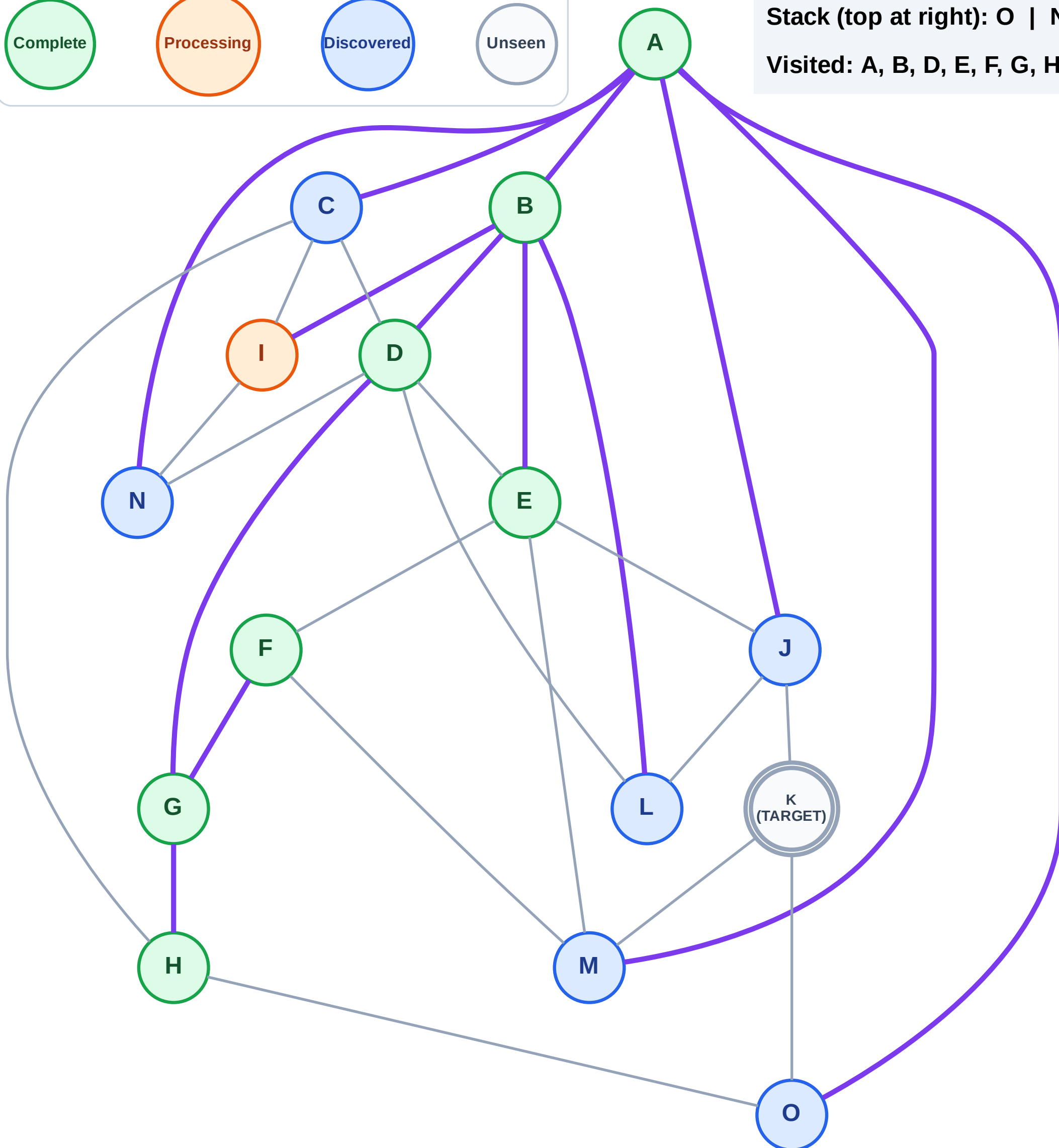
Step 16: Process node I

Legend



Stack (top at right): O | N | M | J | C | L

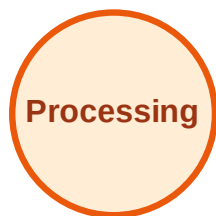
Visited: A, B, D, E, F, G, H



DFS Traversal

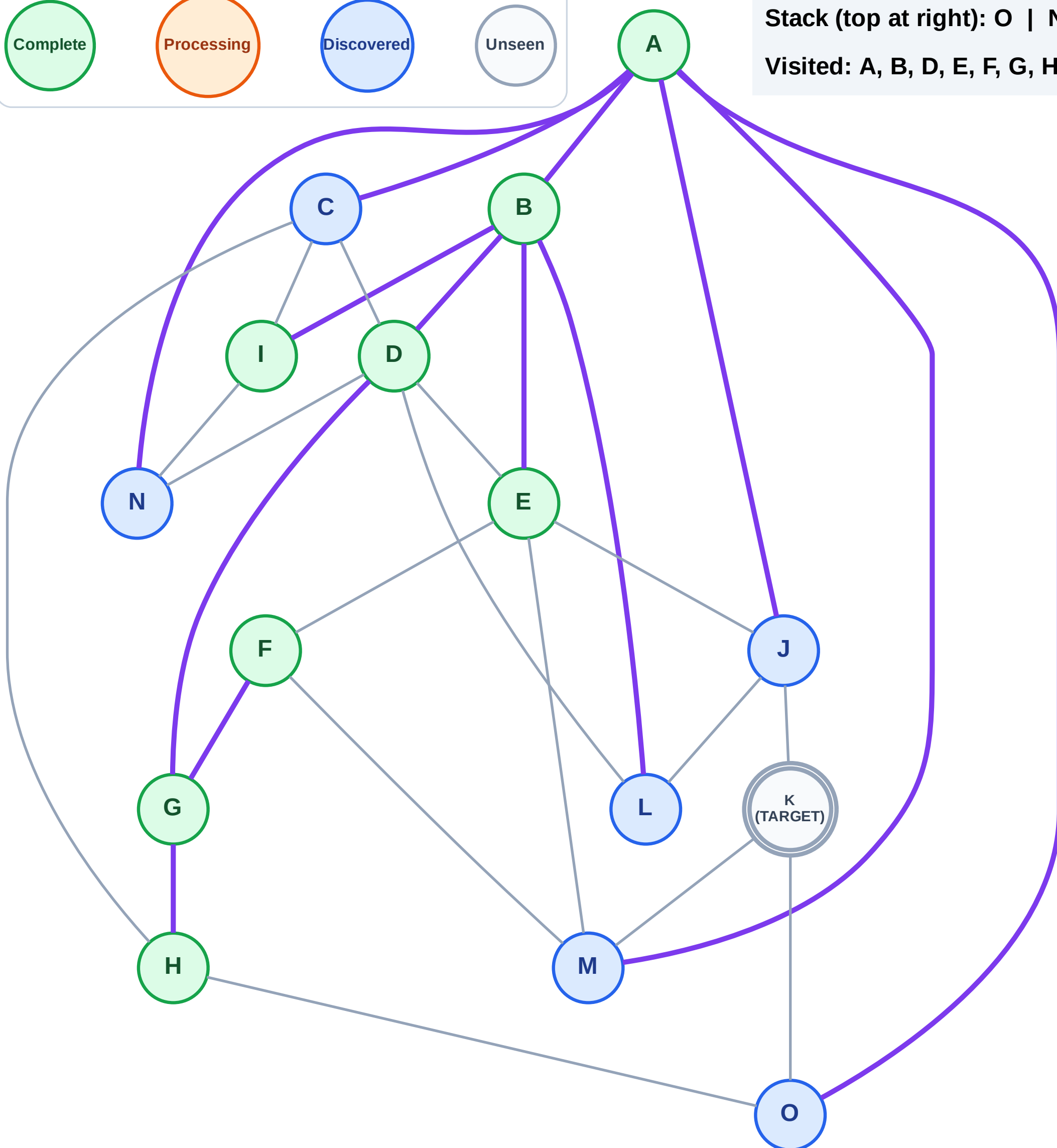
Step 17: No new neighbors from I

Legend



Stack (top at right): O | N | M | J | C | L

Visited: A, B, D, E, F, G, H, I



DFS Traversal

Step 18: Process node L

Legend

Complete

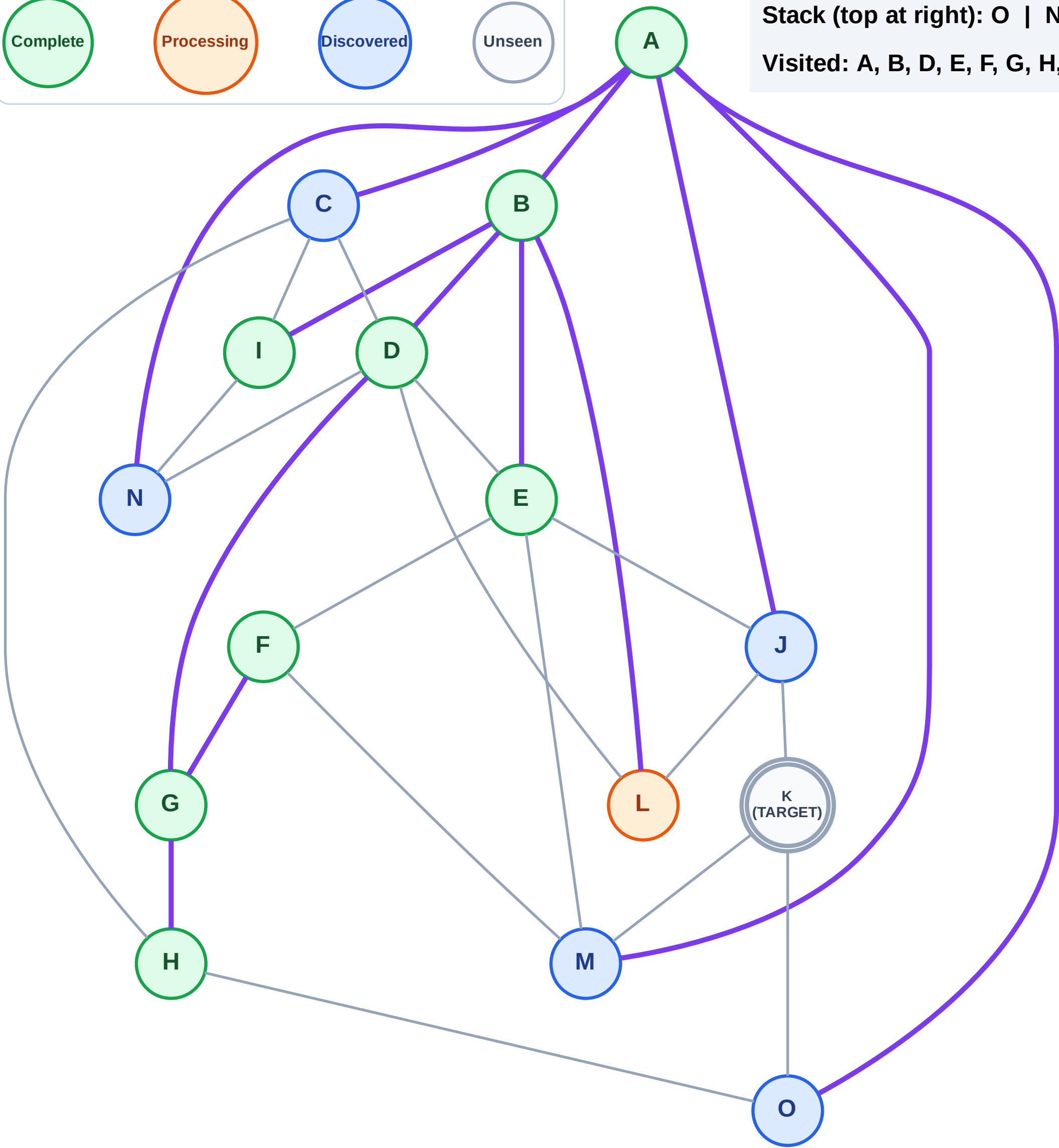
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J | C

Visited: A, B, D, E, F, G, H, I



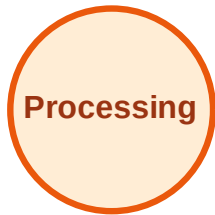
DFS Traversal

Step 19: No new neighbors from L

Legend



Complete



Processing



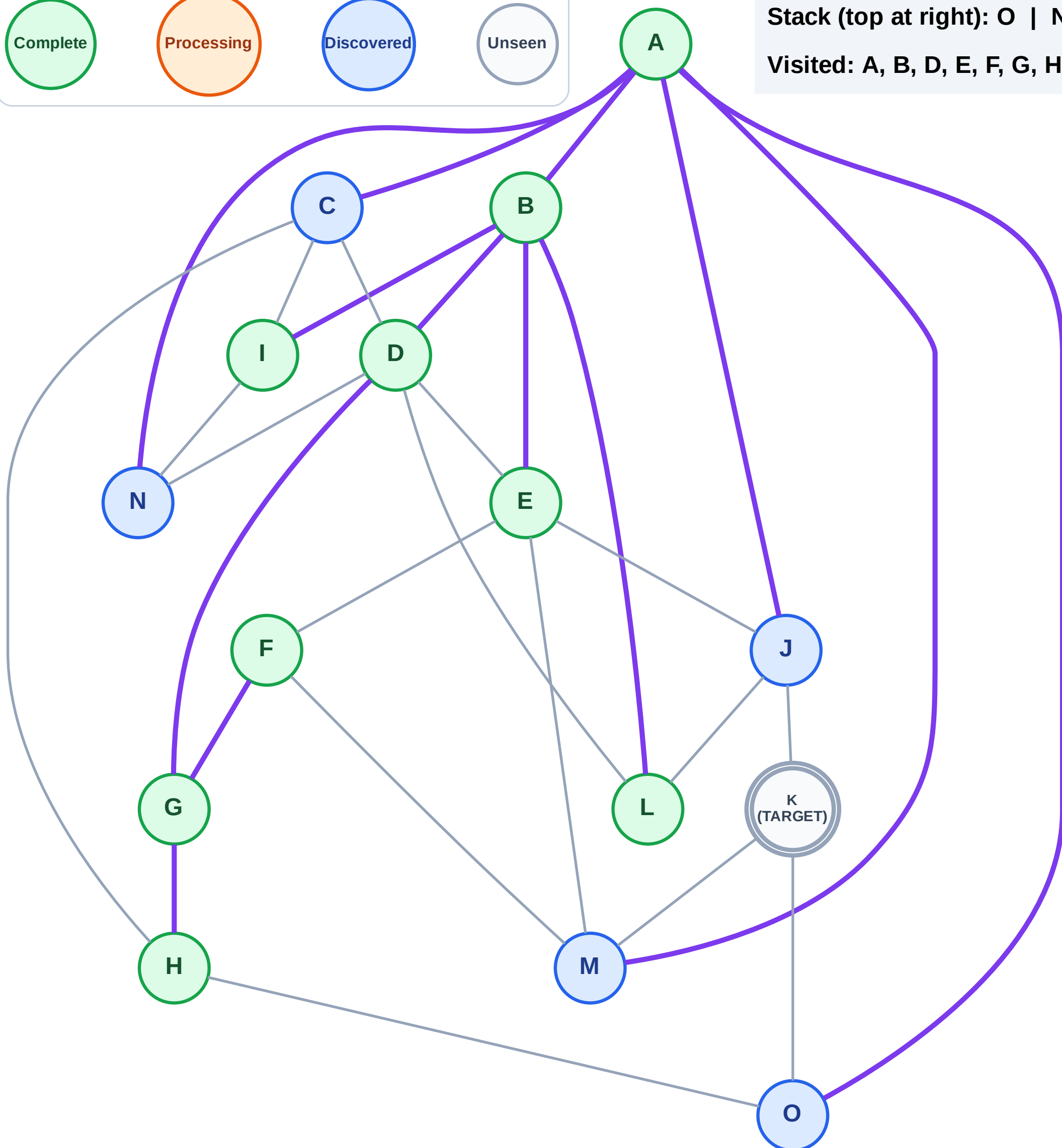
Discovered



Unseen

Stack (top at right): O | N | M | J | C

Visited: A, B, D, E, F, G, H, I, L



DFS Traversal

Step 20: Process node C

Legend

Complete

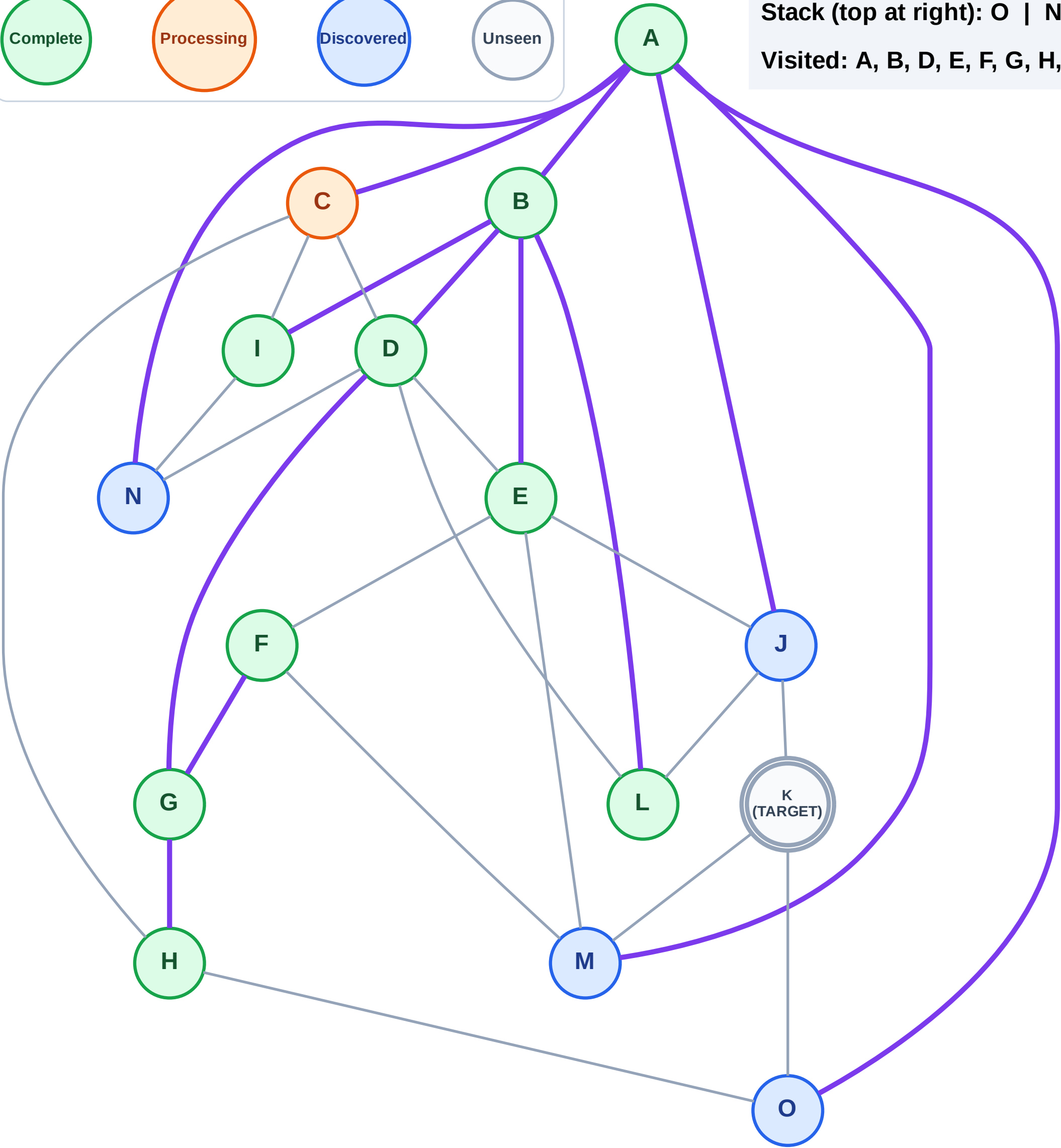
Processing

Discovered

Unseen

Stack (top at right): O | N | M | J

Visited: A, B, D, E, F, G, H, I, L



DFS Traversal

Step 21: No new neighbors from C

Legend



Complete



Processing



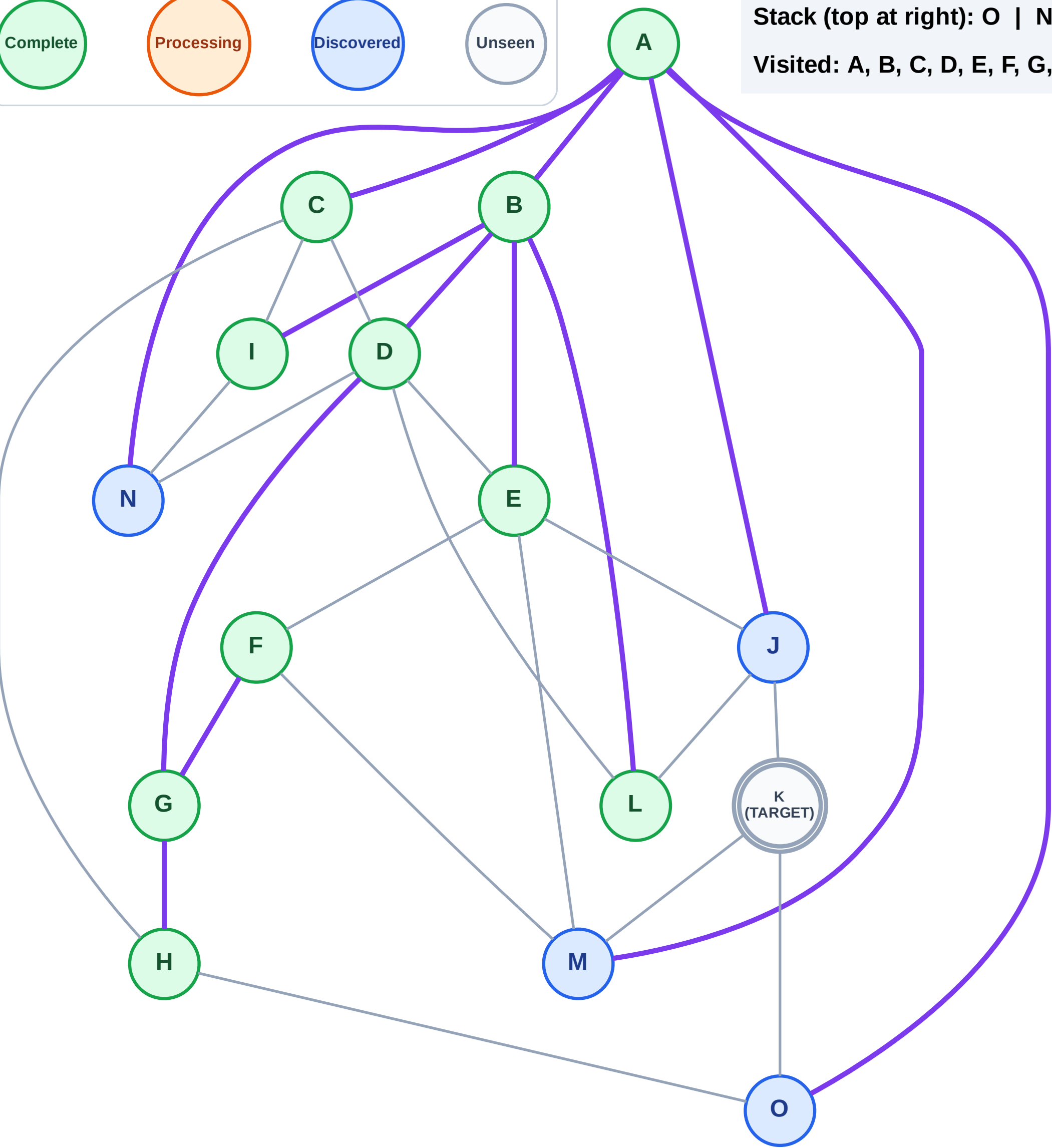
Discovered



Unseen

Stack (top at right): O | N | M | J

Visited: A, B, C, D, E, F, G, H, I, L



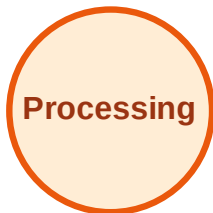
DFS Traversal

Step 22: Process node J

Legend



Complete



Processing



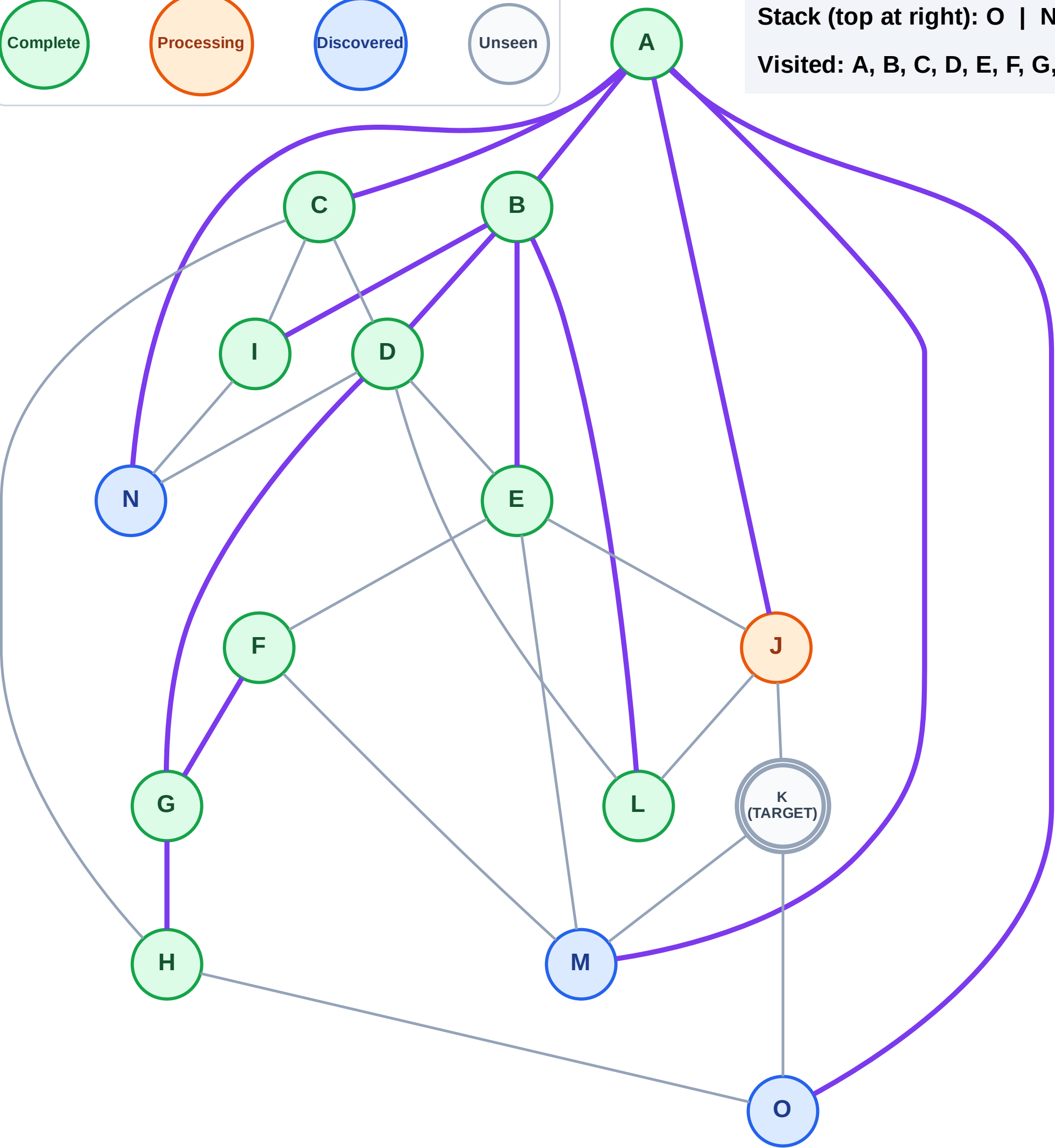
Discovered



Unseen

Stack (top at right): O | N | M

Visited: A, B, C, D, E, F, G, H, I, L



DFS Traversal

Step 23: Discover K from J

Legend

Complete

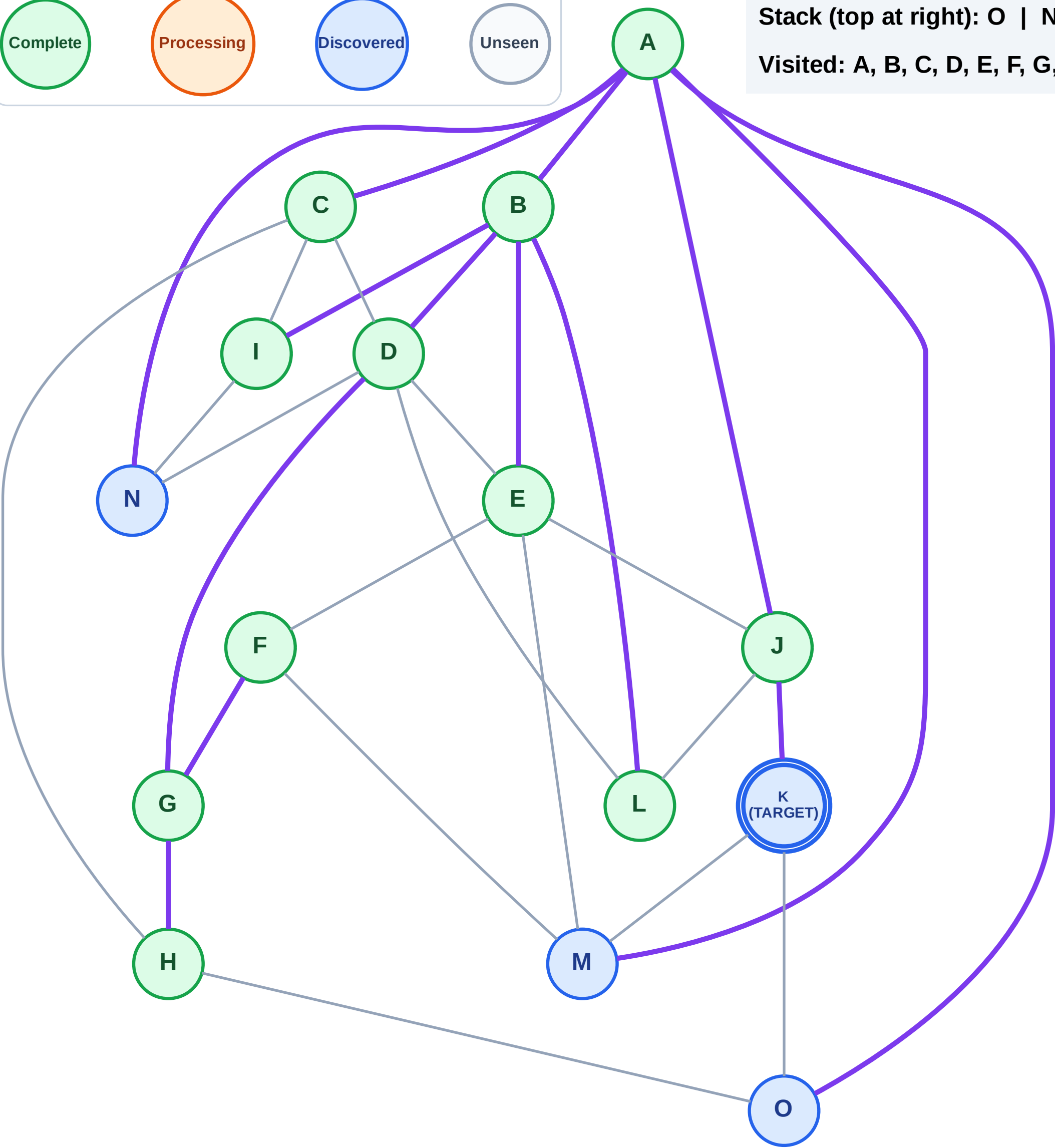
Processing

Discovered

Unseen

Stack (top at right): O | N | M | K

Visited: A, B, C, D, E, F, G, H, I, J, L



DFS Traversal

Step 24: Process node K

Legend

Complete

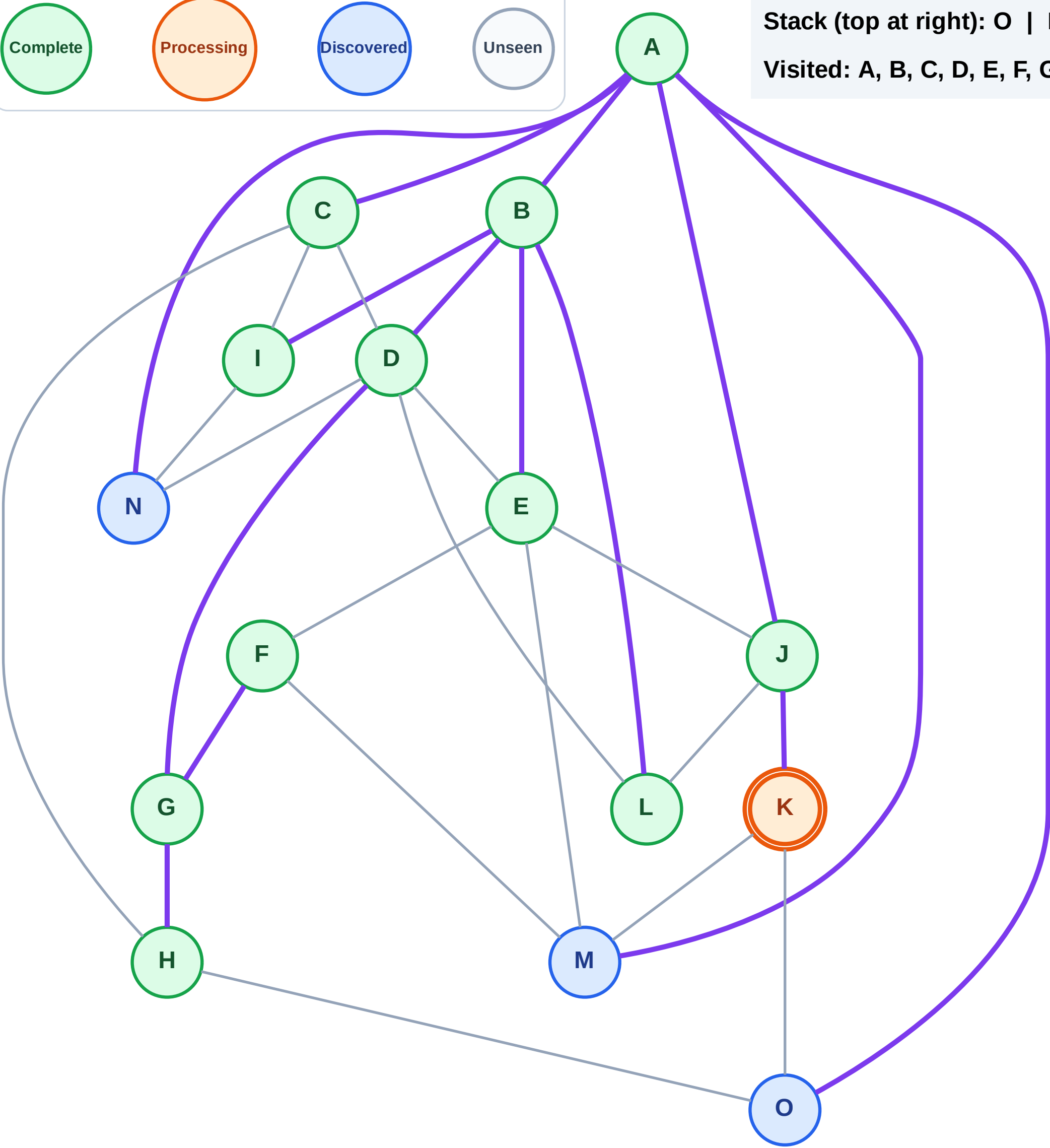
Processing

Discovered

Unseen

Stack (top at right): O | N | M

Visited: A, B, C, D, E, F, G, H, I, J, L



DFS Traversal

Step 25: TARGET NODE K REACHED! Path (2 hops): A → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | M

Visited: A, B, C, D, E, F, G, H, I, J, K, L

